

Feasibility analysis and techno-economic design of grid-isolated hybrid renewable energy system for electrification of agriculture and irrigation area: A case study in Dongola, Sudan

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ABSTRACT

Today, particular attention is being globally paid towards clean and sustainable energy system. The rapid development in renewable energy technologies, growth of energy markets, and adopted financial strategies and policies have the key role in achieving this target. This paper provides a comprehensive feasibility analysis of a grid-isolated hybrid renewable energy system for electrification of agriculture and irrigation area in Dongola, Sudan. A systematic and integrative framework combined with techno-economic optimization analysis for adequate planning and design of hybrid renewable energy system is proposed. Different hybridization cases of a solar photovoltaic, wind turbine, diesel generator, battery storage, and converter technologies, together with a diesel generator-based energy system as base case are modeled, evaluated and compared considering the stochastic behavior of renewable energy resources with a main target to find the most feasible and reliable solution with least system cost and realistic environmental impacts. This followed by a sensitivity analysis to reveal the effects of uncertainty in system parameters on the designed system and its economic performance. The obtained results indicate that solar-wind-diesel-battery-converter hybrid system is of optimal performance and superiority over the studied cases to serve the load demand of the investigated area. The system not only gets the minimum values of the total net present cost (24.16 M\$) and levelized cost of energy (0.387 \$/kWh), but also effectively achieves a positive return on investment of 39.94% and around 95% reduction in both carbon emissions and fuel consumption compared to the base case. Meanwhile, the result of sensitivity analysis demonstrates the high dependability of system cost on the volatility of solar radiation and wind speed as well as the changes in the rate of interest with the prospect of ignoring the effect of variation of diesel fuel price in the future investment decision. Finally, the output of this study is projected to raising the potentiality awareness of renewable energy in Sudan and providing a valuable reference about the optimal planning and utilization of hybrid renewable energy system in energy sector.

1. Introduction

In recent decades, increasing people populations and higher technology evolution causes rapid growth in electrical energy demand. Total energy demand is expected to increase by around 30% reaching to 2040 compared to 2015 [1]. Nowadays, approximately 75% of global electrical energy production comes from fossil fuel-based energy sources (named conventional energy sources (CESs) [2]. Besides the anticipated shortage in fossil fuels caused by the heavy decrease of their natural reserve [3] and progressive increase of its price in the upcoming

years [4], conventional resources are the main factor of increasing greenhouse gas (GHG) emissions [5]. To add with, there are a significant number of population (~1.1 billion people) still live with lack get electricity [6], particularly in developing nations [7]. This alarming situation urged utilities and policymakers to establish alternative solutions in order to enforce the environment preservation and provide universal and sustainable electricity of all [8].

To serve this purpose, Renewable Energy Sources (RESs) [9] and energy efficiency [10] have been recognized globally as a panacea and an optimal and efficient pathway to overcoming the last mentioned

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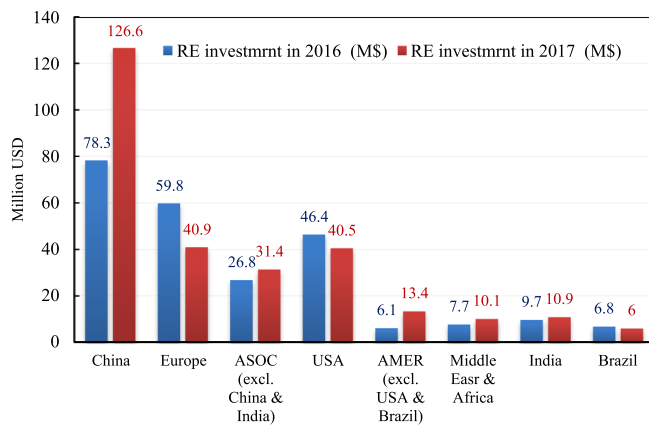


Fig. 1. Renewable energy investments in 2016 and 2017 categorized by region [11].

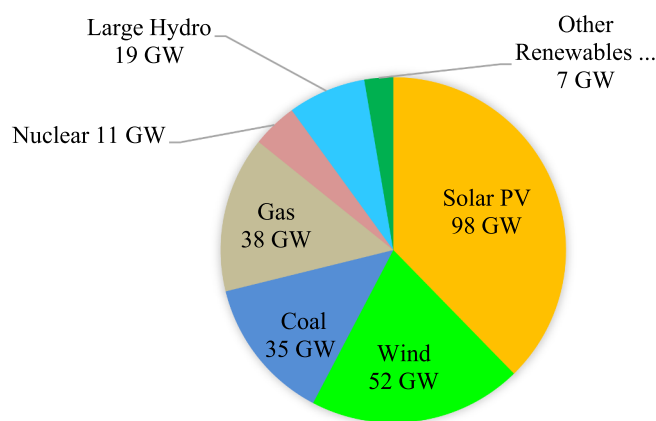


Fig. 2. Globally added power by renewable technologies, 2017 [12].

issues associated with CESSs. Furthermore, RE can share effectively in enhancing the socio-economic developments, especially in developing countries by increasing the national income and offering new job opportunities. In 2016 around 9.8 million jobs were provided by different RE technologies [2]. As a result, the global new investments in RE technologies dramatically increase year after year. Fig. 1 present a comparison between global renewable energy investment in 2016 and 2017 [11]. Three biggest developed countries 'China, Brazil and India' have been considered as the energy-starved nations and have recorded an extensive increase in renewables investment from 94.8 M\$ (54% of the global investment) in 2016–197.5 M\$ (63% of the global investment) in 2017. On the other side, the share of the European region in 2017 has reduced to the lowest value of 40.9 M\$ (15%) since 2014. The provided pie chart in Fig. 2 demonstrates that around 176 GW produced from various renewable technologies (solar: 98 GW, wind: 52 GW, others: 26GW) have been added to the global power capacity by 2017 with an average increase by 10% per year since 2010 [12]. In contrast, only 70 GW has offered by conventional resources are added to the global power capacity in the same year.

Motivated by the worldwide rapid progress in RE technologies, particularly PV and WT, growth of new energy markets, and enhancement of financial strategies adopted by policies makers [13], the global Renewable Energy roadmap to 2050 recognized by international renewable energy agency (IRENA) has planned a goal of expanding the cost-effective share of renewables in the global energy generation to 85% by 2050 compared to 25% in 2017 [2]. Accordingly, renewables can efficiently match the United Nations Sustainable Development Goals (UNSDGs) and Paris Historic's Climate Agreement (HCA) that were put in place in 2015. Worth to mention that HCA was established

with the agreement of 195 nations aiming at regulating the global temperature rise well below 2 °C in the present century by decreasing the total CO₂ emissions to 470 Gt reaching 2050 [2].

Despite the enormous benefits offered by RESs, they bring unreliability and low-security issues when individually used to supply load demand. This main reason behind that is the intermittent and stochastic nature of the RE resources, where the generated power from RESs substantially depends on the variable climate conditions [14] (i.e., solar irradiation, average sun hours, wind availability and speed, etc.). Energy storage systems (ESSs) penetration can subsidize to the retention of a constant power supply and load leveling [15], but their high initial cost and energy loss contribute significantly to increase the overall system cost and reduce energy conversion efficiency [9]. In this context, Hybrid Renewable Energy Systems (HRESs) were developed as a new dawn, suitable green and cost-effective solution for most of the technical, financial and environmental problems associated with electricity generation. HRES is defined as a combination of RESs (e.g., photovoltaic (PV), wind turbine (WT), fuel cell (FC), hydro) and CESSs (e.g., diesel, micro-turbine) equipped with ESS, used to meet load demand (e.g., electrical, thermal, hydrogen) while optimizing the system economy, atmospheric emissions and supply reliability, either in grid-connected or grid-isolated architecture [16]. To achieve the utmost benefits from HRES, the system must be well designed and planned, and it will be the main concern of this paper.

Over the past decade, many researches have been developed to investigate and review the topic of HRE-MG systems from several aspects; planning and design [17], control and management [12], and optimization [18]. They considered various combinations of RESs, CESSs, and energy storage system in both on-grid [19] and off-grid [20] modes. According to the scope of this paper, the planning and design of grid-isolated HRES are reviewed and evaluated as given in the literature. In this sense, a comprehensive survey on various case studies of HRES focus on grid-isolated systems, which has been implemented in various locations all over the world is performed and summarized in Table 1. The table lists the modeled HRE technologies, load type, and corresponding energy consumption, optimization criteria for optimal planning and design, sensitivity variables, and optimized financial results as well. Besides, the main findings of each study are highlighted in the last column of the table.

The main highlights from the literature are (i) the HRESs are perceived as the most affordable and reliable energy source compared to CESSs in many locations all over the world; (ii) there are many configurations of HRES that can be implemented to meet the local demand, in which, the selection of the proper combination of hybrid RE technologies varies according to the meteorological features of the studied area (weather conditions and potential of RE resources), load requirements, rapid development in RE technology and financial and environmental strategies; (iii) most of the conducted studies have evaluated the financial and technical practicality of HRES for electrification of rural and urban communities, neglecting the energy requirement by agricultural and irrigation sector; and finally (v) to the best of the author's knowledge, there is a lack of studies that provide a strategic context and integrated approach for the optimal planning and further development of HRESs, regardless of the growth of energy markets and numerous adopted financial policies that lead to rapid utilization of HRESs [58].

Based on the above consideration, the scope of this work are:

- To provide a systematic and integrated framework for optimal planning and design of HRES. The primary purpose of this framework is to create a straightforward and strategic approach to differentiate between different key elements related to HRES, and also to clarify an adequate procedure that ensures, as possible, the optimal planning and design as well as assessment of the hybrid energy systems.
- To apply the proposed framework in order to study the feasibility and assess the performance of PV-WT-DG-battery-converter HRES to

Table 1
Summary of grid-isolated HRES in the literature.

Ref	Location	HRE technologies	Energy consumption/Load category	Optimization criteria	Sensitivity variables	Optimized cost results		Findings
						NPC	COE	
[21]	Saint Catherine, Egypt	PV/WT/FC/DG/Natural gas turbine/Biomass generator/Battery/Converter	80 kWh/day Residential load in a rural village	E/V	Load demand	165,157 \$	0.532 \$/kWh	Developed an approach for optimal sizing of HRES using bi-directional ant colony algorithm with objectives of minimizing both NPC and CHG emissions, simultaneously
[22]	Shafar, Yemen	PV/WT/DG/Battery/Converter	886 kWh/day Typical 50 house hold load in rural area.	E/V	Solar radiation Wind speed DG fuel price	722,356 \$	0.137 \$/kWh	Current and future feasibility of hybrid PV/WT/diesel hybrid system over 100% RE (i.e. PV/WT) and diesel systems
[23]	Hurghada, Egypt	PV/WT/DG/Battery/Converter	1153 kWh/day, Residential load	E	DG fuel price	822,578 \$	0.139 \$/kWh	Economic and environmental benefits of using HRES deployment instead of diesel only system for electrification of the investigated load demand
[24]	India	Biomass generator/PV/WT/Battery/Converter	110.6 kWh/day Residential load	T/E/V/S	–	**	0.195 \$/kWh	Formulated a new multi-objective function with six objectives (COE, LPSP, RF, PM, HDI, JCI & GHG) for optimal planning of diverse HRESs. Problem solved using different optimization techniques (GA, BPPSO, PSO and TLBO). The investigation indicated that that TLBO as the powerful tool to handle all problem objectives and give an optimal solution
[25]	Shanghai, China	PV/WT/Battery/Converter	255 kWh/day Residential load at a remote island	T/E	Load demand	**	0.0943 \$/kWh	Developed an adequate approach for dimensioning a grid isolated HRE-system considering the effect of increasing the saturation level of one of the involved RE power sources meanwhile decreasing the ratio of another source on the system design stage
[26]	Hargeisa, Somalia	PV/WT/Battery/Converter	1283 kWh/day Residential load in an urban center	E	Solar radiation Wind speed WT O&M cost DG fuel price Electrical load	1,720,000 \$	0.288 \$/kWh	Investigate the possibility of using HRES for supplying the studied urban area considering the future increase in load demand
[27]	Tanzania	PV/DG/Battery/Converter	1.55 kWh/day Residential load	E	DG size Load demand	**	1.107 \$/kWh	Studied the economic viability of the implementation of an isolated microgrid system as a backup in the case of a national grid blackout to meet the load demand. Mention that the solution is based mainly on blackout duration and level of load demand
[28]	Silchar, India	PV/biogas generator/pumped hydro energy storage/Battery/Converter	356.38 kWh/day Radio transmitter station	T/E	LPSP design factor	813,319 \$	0.4864 \$/kWh	Solved the problem of the techno-economic optimal design of an off-grid HRES using a new metaheuristic optimization technique, namely MFO and WCA
[29]	Yuxi, China	PV/FC/Retired electrical vehicle battery/Converter/Electrolyzer	**Residential load	T/E	–	825,897 \$	**	Formulated a robust mathematical model for a retired electric vehicle battery. Provide a tri-objective optimization model to find the optimal HRES design
[30]	Cesme, Izmir, Turkey	PV/WT/FC/Battery/Hydrogen tank/Converter/Electrolyzer	165.59 kWh/day (regular occupancy and 100.61 kWh/day (seasonal occupancy)	E	–	**	0.186 \$/kWh	Investigated the techno-economic viability of hypothetical off-grid HRES under two options for energy storage (battery and hydrogen) to meet the electrical energy demand for the coastal area considering different load profiles for seasonal and regular occupancy. Concluded the economic-wise of battery energy storage system over hydrogen with all studied scenarios
[31]	Napier, Western Cape, South Africa	PV/WT/FC/Battery/Hydrogen tank/Converter/Electrolyzer	1080.6 kWh/day farming village	E	hydrogen cost	107,000,000 \$	21.02 \$/kWh	Presented a comparative study between grid-connected and grid isolated HRES
[32]	Shiraz, Iran	PV/WT/DG/Battery/Converter	9911 kWh/day Large non-residential load	E	Electrical load Interest rate Carbon tax	**	0.126 \$/kWh	Evaluate the effect of different techno-economic parameters on the hybrid system in both stand-alone and grid-connected modes
[33]	South China Sea, Malaysia	PV/WT/DG/Battery/Converter	9126.49 kWh/day Large resort centre, tourist sectors	E	Solar radiation, Wind speed DG fuel price	17,150,000 \$	0.279 \$/kWh	Focused on tourist sectors and give feasible HRES solution for electrification of the investigated load demand

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Table 1 (continued)

Ref	Location	HRE technologies	Energy consumption/Load category	Optimization criteria	Sensitivity variables	Optimized cost results		Findings
						NPC	COE	
[34]	Manit, Bhopal, India	Biomass gasifier generator/PV/FC/Battery/Converter	101 kWh/day Energy center	E	–	5,189,000 Rs	15.06 Rs/kWh	Suggested the optimal size of HRES for electrification of the investigated load
[35]	Sundarban region, India	PV/WT/DG/Battery/Converter	5.755 kWh/day Residential load with typical 15 houses	T/E	PV, WT and DG capacities	**	62.27 Rs/kWh	Find the best hybrid system configuration and component size based on minimum COE and loss of power supply (LPS) and maximum RF
[36]	Oujda, Morocco	PV/WT/Battery/Converter	18.7 kW/day Residential area	T/E	–	37,878 €	0.3747 €/kWh	Presented a new algorithm for optimal designing of stand-alone HRES based on Modified Electric System Cascade Analysis method
[37]	KSA	PV/WT/Battery/Hydrogen tank/Converter/Electrolyzer	**Residential load	E	–	38,523\$	0.609 \$/kWh	Design cost-effective renewable generation systems for power generation and hydrogen production in different climate conditions. Highlight the possibility of replacing battery storage with FC, hydrogen tank and electrolyzer
[38]	Tindouf, Algeria	PV/WT/DG/Battery/Converter	82.14 kWh/day Sahrawi refugee camps	E/S	–	125,465 €	0.21 €/kWh	Developed a multi-objective evolutionary algorithm to find the most feasible solution of a stand-alone HRES to meet load demand of the investigated area. Considered minimization of NPC and maximization of both HDI and JC in the proposed objective function
[39]	Puerto Estrella, Colombia	PV/WT/DG/Battery/Converter	**Residential load in rural area	E	–	836,210 \$	0.473 \$/kWh	Analyzed the application of HRES for rural electrification with different climatic characteristics through a techno-economic feasibility study. Considered NPC, COE and initial capital cost as economic indicators, and CO2 as an environmental index
[40]	Chorasariadho, Bangladesh	PV/WT/DG/Battery/Converter	242.56 kWh/day Village community with 220 households	E/V	Discount rate PV cost DG cost Battery cost Fuel cost Capital costs	335,000 \$	0.37 \$/kWh	Presented an optimized off-grid HRE compared to grid-connected and solar home systems. Analyzed different perspectives related to the technical, economic, environmental, and social performance of a hybridized energy system
[41]	Brookhaven, USA	PV/DG	518 kWh/day Commercial load	T/E/V	–	**	**	Present comprehensive technical and economic analyses regarding the optimal location, size, and operation of HRES system for supplying the studied area
[42]	Rafha, KSA	PV/DG/Battery/Converter	44 kWh/day Residential load	E	–	28,500,000 \$	0.170 \$/kWh	Assessed the techno-economic feasibility of the hybrid system for a typical remote village and examined the variation of COE, diesel operating hours and carbon emissions with different PV/battery capacities
[43]	Gwagwalada, Abuja, Nigeria	PV/WT/DG/Battery/Converter	12.51 kWh/day Residential load, rural communities	T/E	Load demand	82,283 \$	0.3145 \$/kWh	Battery performance and reliability are evaluated for different configurations of hybrid systems with increasing load demand
[44]	Amritsar, India	PV/WT/DG/Battery/Converter	1.3 kW* Telecommunication service	E	–	10,015 \$	0.164 \$/kWh	Present PV/WT/diesel hybrid system as the most feasible as a solution for supplying the load with minimum COE and maximum power generation
[45]	Almora district of Uttarakhand state, India.	Micro-hydro/Biomass diesel/Biogas diesel/PV/WT/Battery/Converter	1537.7 Residential load with 267 households	T/E	Solar radiation Load demand Fuel cost	**	3.36 Rs/kWh	Analyzed the economic and technical aspects of different scenarios of an integrated RE system to meet the load demand with least COE and optimum system reliability parameters (i.e. EEI and EENS)
[46]	Lanzarote, Canary Islands	PV/DG/Battery/Converter	250 kW h/day Desalination water system	E	Wind speed Load demand	473,013 \$	0.404 \$/kWh	Proposed and analyzed a standard HRES to guarantee the power requirements of an autonomous desalination system from the technical and economic perspective
[47]	Pulau Banggi, Sabah, Malaysia	PV/WT/DG/Battery/Converter	6632.35 kWh/day Rural area	E	Fuel price PV capital price battery price Demand load	8,540,000 \$	0.276 \$/kWh	Analyzing the operational behavior of an optimized HRES using HOMER for electrification of the investigated area as the best alternative for a standalone DG system and an existing HRES (which was non-optimized system)

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Table 1 (continued)

Ref	Location	HRE technologies	Energy consumption/Load category	Optimization criteria	Sensitivity variables	Optimized cost results		Findings
						NPC	COE	
[48]	Adrar, Algeria	PV/WT/DG/Battery/Converter	125 kW*	E/V	-	**	**	Designed a proper HRES for supplying the investigated area with minimum life cycle cost
[49]	Yanbu, KSA	PV/WT/DG/Battery/Converter	**Single household load	T/E	PV, WT and DG penetration levels	**	0.081 \$/kWh	Develop multi-objective (i.e. COE and LPSP) optimization approach for optimal scheduling of hybrid energy system components
[50]	Kapit, Sarawak, Malaysia	PV/DG/FC/WT/Battery/Converter/Electrolyzer	140.75 Residential load with 50 households	E/V	DG fuel price Load demand	335,297 \$	0.323 \$/kWh	Studied the feasibility of using HRES in the site location and suggested the most economical and environmentally green configuration
[51]	Bishopton, Scotland	PV/WT/DG/Battery/Converter	15 kWh/day Residential load with typical 3 houses	E	Demand load	**	0.677 £/kWh	Design a new tool for modeling and optimal sizing of HRES. Use minimum life-cycle cost as a decision criterion and explore the trade-off between the battery and diesel sizing
[52]	Layyah, Punjab, Pakistan	PV/Biomass/Battery/Converter	168.36 kWh/day Agricultural farm in a remote rural area	E	Biomass amount Solar irradiance	4,480,000 Rs	5.51 Rs/kWh	Assessed the techno-economic feasibility of the hybrid system for agricultural purposes
[53]	Selangor, Malaysia	PV/WT/DG/Battery/Converter	33 Small residential community	E/V		288,000 \$	1.877 \$/kWh	Proposed a hybridization of PV/WT/diesel/battery as the most economical and environmentally efficient system for supplying the investigated area instead of the diesel system
[54]	Kerman, Iran	PV/FC/DG/Converter/Electrolyzer	off-grid rural community**	T/E	Operating reserve Solar radiation Load demand	1,241,000 \$	**	Developed MOCSA to find the optimal size of HRES components considering minimization of NPC and LPSP as the optimization objectives. It is concluded that the integration of the hydrogen system will provide more reduction in the overall cost of the HRES
[55]	Rafsanjan, Iran	PV/WT/Battery/Converter		E	-	17934.79\$	**	Proposed a methodology for the designing of off-grid HRES by taking into account the uncertainty in solar radiation, wind speed, and load demand using PSO combined with Monte Carlo simulation method
[56]	Busan, South Korea	PV/WT/Battery/Converter	Metropolitan city	E/V	-	26,090,000 \$	0.399 \$/kWh	Formulated the optimal HRES configuration 1/500 scaled city load based on minimum NPC and COE values
[57]	Vancouver, Canada	PV/DG/Natural gas generator/Biomass/Battery/Converter	Urban community (residential, commercial, and industrial load patterns)	E/V		48,000,000 \$ (commercial)	0.399 M\$/kWh (commercial)	Investigated in depth the economic and environmental performance of three power scenarios (i) 100% natural gas; (ii) hybrid NG/renewables; (iii) 100% renewables for residential, commercial, and industrial load sectors. Show the profitability of biomass gasifier to reduce the system cost

* Peak power demand, ** Unknown value, E: Economic, S: Social, T: Technical, V: Environmental.

BFPSP: Butterfly Particle Swarm Optimization, COE: Cost of Energy, EEI: Energy Index Ratio, EENS: Expected Energy Not Supplied, GA: Genetic Algorithm, GHG: Green House Gas, HDI: Human Development Index, JCI: Job Creation Index, LPSP: Loss of Power Supply Probability, NPC: Net Present Cost, PM: Particular Matter, PSO: Particle Swarm Optimization, TLBO: Teaching-Learning-Based Optimization, MFO: Moth flame optimization, WCA: Water-cycle algorithm.

meet the energy requirements for agriculture and irrigation area located in Dongola, Sudan, where the utility-grid was not available.

- To investigate the techno-economic optimized configuration plan of the hybrid system that will be able to satisfy the load demand with high reliability level (means near zero capacity shortage) under main objective of a minimizing the total NPC, as well as reasonable environmental impacts by analyzing, assessing, and comparing the techno-economic and emission performance of different hybridization scenarios.
- To analyze the financial performance of the designed HRES under critical conditions of uncertainty regarding RE resources and variations of fuel price and interest rate.

The remainder of this work is arranged as follows, the proposed framework of HRES planning and design, along with the techno-economic modeling and optimization of the hybrid energy system is introduced in Section 2. RE prospect of the investigated area in Sudan and case study data are presented in Section 3. Study results and further discussion are reported in Section 4. In the end, the conclusion and research findings are presented in Section 5.

2. Material and methods

In order to obtain the optimal design of HRES and adequately evaluate the system performance, firstly a generic and integrated framework is developed as a decision-support tool to help the system planners to conceptualize the most appropriate solution. Because of the techno-economic optimization analysis is an essential step in the proposed framework, so it is necessary to develop the technical, economic and carbon emissions models of HRES. Subsequently, by the optimizing the objective function (which is minimizing the total NPC), subject to a set of design constraints, the optimal configuration plan of the system components will be obtained. Then, system performance can be assessed in terms of its techno-economic and environmental aspects.

2.1. Proposed framework for optimal planning and design of hybrid renewable energy system

According to literature, several authors have studied different configurations for the HRESs in various location worldwide. However, they did not identify a clear strategic context for hybrid energy systems planning and design. Therefore, a systematic framework was proposed in this study. The main purpose of this approach is to create an integrative and compelling case between different key elements related to HRES, and also to clarify adequate procedure that ensures as possible the optimal planning and design as well as assessment of the examined system. So, the proposed framework act as a decision making approach that assists in optimal utilization and adaptation of HRES as the best pathway to ensure sustainable and clean energy sources. The framework consists of five main stages presented in Fig. 3.

First of all, the drivers behind HRE investment implementation is necessary to be well-defined. This can justify the business case of the system. The proposed framework provides a synopsis of the main drives behind the implementation of HRES which include:

- Technical drives: a well-designed HRES can provide diversified technical benefits for utilities and HRES members such as removing reliability issue and ensuring high power quality;
- Financial risk: HRE have usually implemented close the load; thus it offers favorable incentives to relieve the energy sector investors from the additional budgets associated with setting up of conventional centralized systems, transmission and distribution lines, etc.;
- Environmental policies: aiming at promoting HRES to combating climate change and decreasing the harmful CO₂ emissions for HRES members;
- Technological developments associated with RE and energy storage:

with the recent developments and vast progress in the utilization of RE and energy storage, HRES can be recognized as a cost-effective solution in the new energy market; and

- Regulatory schemes: target to promote the diversity of power sources to strengthen energy security.

In the second process, a preliminary feasibility analysis should be involved. This will help in selecting the candidate HRE technologies and create a proper synthesis between them to fit the local load demand. In fact, this step is suggested to be based on a foresight approach. Accordingly, it is required to evaluate the key inputs of HRE planning and design, which are:

- Meteorological data and availability of RE resources in the site location;
- Estimated daily and seasonal load demand profile and its categories, which depend mainly on the type of inhabitant;
- Current power generation system supplying load demand; and
- Energy system constraints associated with system designing that can limit or excludes the implementation of some energy technologies such as the geographical restraints (e.g., only a specific area is available for PV panels installation or issues that obstructed the stability of WT structure).

Once the candidate energy technologies are adopted, the potential optimal plan needs to be determined. For any optimization problem, the choice of the optimization objective or criterion is governed by the nature of the problem. Considering that HRESs typically have significantly various cost characteristics and lifetimes [59]. Also, incorporating environmental criteria is less straightforward and less certain than a performance or cost-based assessment [60]. So, technical and economic factors are mostly folded into the optimization framework associated with the hybrid energy system as the critical investigator to select the best adequate solution. Consequently, it is crucial to provide an inclusive techno-economic simulation and optimization analysis as mentioned in the third stage of the proposed approach (stage C). This process has to be appropriately performed through developing a correct model of technical and economic parameters of the available options of energy technologies.

Additionally, to avoid wrong sizing of the system component, it is crucial to maintain the right operational and optimization strategy along with the trustworthy economic investigator. Following this stage, it is intuitively that the proposed optimized HRES plan need to be assessed regarding its operational, economic and environmental characteristics. This process will assist in quantifying the outcomes attained by the designed system. The proposed framework outlines how the designed system can be evaluated from different perspectives in the fourth stage (stage D).

After completing the system optimal planning and assessment, it is essential to perform sensitivity analysis. This study is required to measure the financial sensitivity of the energy system against the uncertainty of system parameters, which are beyond the designer's control. It is also essential to understand how the future investment case could be managed and to answer a general question about technologies options to inform planning and policy decisions. The way of implementation of techno-economic optimization, as well as sensitivity analyses, are addressed in the upcoming sections.

2.2. Techno-economic simulation and optimization analysis

In order to explore the techno-economic feasibility of HRES configurations within optimum system dimensions, different evaluation criteria have been considered in the literature. They were developed usually based on specific economic factors such as NPC (or life-cycle cost), COE (or levelized cost of energy), total system cost, system annualized cost, average generation cost, etc. [17]. While RE technologies

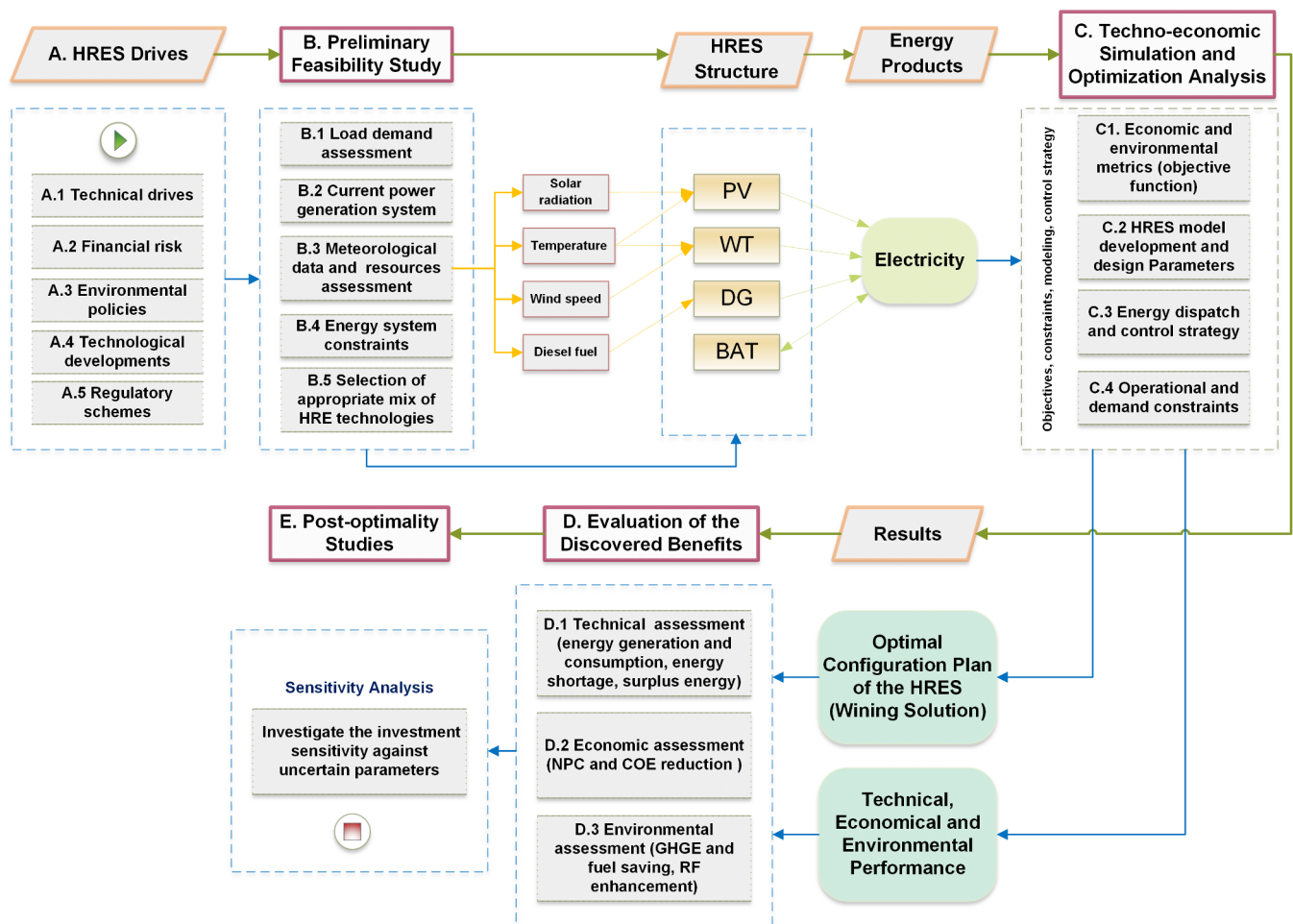


Fig. 3. Proposed systematic framework for optimal planning and design of HRES.

involve higher initial capital cost and relatively free O&M cost, the inclusion of diesel and battery will considerably share in increasing system O&M cost (e.g., fuel consumption rate and batteries replacement). Therefore, assessment of the system's economy based on total life-cycle cost will be a more trustworthy metric and lend credibility to the optimized design as a defensible concession. In this sense, minimization of total system life-cycle cost was used as the system optimization objective function to find the winning configuration plan of among various energy system alternatives concerning technical, economic and environmental constraints. Conclusively, the HRES elements as decision variables (e.g., Solar PV arrays size, rating of diesel generator, number of WTs, power converters, and batteries packs) were optimized under the main condition of satisfying the demand load with zero capacity shortage as well as other system design constraints under minimum NPC as a main optimization objective (see Fig. 4). Also, the optimal HRES configuration plan was assessed regarding its operational performance, atmospheric GHG emissions, repetition of the total load demand between the adopted energy sources. A description of the design specifications and mathematical technical, economic, and emissions models of the developed HRES are presented in the following sections.

2.2.1. Design specifications and technical model development

The general structure of the proposed hybrid energy system consists of a solar PV array, wind turbine, two diesel generators, battery storage system, and power converter. Fig. 5 show the overall schematic of the proposed HRES. Following the proposed framework discussed above, these components were selected as preliminary feasible technologies

regarding the presented case study. In general, the study area has high availability of solar and wind resources; thus, solar PV array, wind turbine were considered as the best options and efficient RE technologies to be installed in the site location. Also, a diesel generator was regarded as a backup source to cover the excessive energy shortage situations regardless of its harmful emissions impact and high operating and maintenance cost. A battery system is included as a crucial part of the HRES to maintain the power balance between generation and consumption and maximize the exploitation of RESs. Finally, the power converter was added basically for the energy conversion process. The next subsections depict the model components, which are used to simulate the HRES including their technical and economic parameters.

2.2.1.1. Solar photovoltaic. PV has used to converts solar irradiance into DC electricity which evacuated in the AC system via system converter. MaxPower CS6X-325P PV flat plate model was used in this study with a rated capacity of 0.325 W and efficiency of 16.94% and a lifetime of 20 years. A value of 88% site derating factor, as well as ambient temperature effects, were taken into account where the nominal operating cell temperature (NOCT) is 45 °C. The PV panel is horizontally fixed at slope 19.20° and operate without maximum power point tracking system. Capital and replacement costs were considered the same with 700 \$/kW, and O&M cost is 14 \$/kW/year [30]. In this study, worth to declare that the upper bound of the capacity of the PV arrays was capped at 2000 kW. This due to some limitations causes by the topographical characteristics of the investigated area, whereas around 55 km² was estimated to be available for PV arrays installation based on PV panel dimensions and topographical

constraints of the site location. Under the utilized circumstances, the supplied power of the PV panels considering the effect of temperature was calculated as follows [57]:

$$P_{PV} = P_{PV,r} \times f_{PV} \times (H_T/H_{T,STC}) \times [1 + \alpha_p(T_C - T_{C,STC})] \quad (1)$$

where, $P_{PV,r}$ is the rated capacity or power output of the PV array under standard test condition (kW), f_{PV} is the PV derating factor, α_p is the temperature coefficient of power (%/°C), and H_T is the solar radiation incident on the PV array at actual conditions, while $H_{T,STC}$ is the solar radiation incident at the standard test conditions (STC) (800 W/m²). Also, T_C and $T_{C,STC}$ are, respectively, the real-time and STC of the PV cell temperatures (°C), where the real-time PV cell temperature can be calculated from the following equations [61]:

$$T_C = T_a + (T_{C,NOCT} - T_{a,NOCT}) \times (H_T/H_{T,NOCT}) \times [1 - (\eta_{mp}/\tau\alpha)] \quad (2)$$

$$\eta_{mp} = \eta_{mp,STC} \times [1 + \alpha_p \times (T_C/T_{C,STC})] \quad (3)$$

where, T_a is the ambient air temperature (°C), $T_{C,NOCT}$ is the nominal operating cell temperature (45 °C in this study), $T_{a,NOCT}$ is the ambient temperature at which $T_{C,NOCT}$ is calculated (°C), $T_{C,STC}$ is the PV cell temperature at STC (25 °C) and η_{mp} , $\eta_{mp,STC}$ are maximum power point efficiencies of the PV array at real time and STC, respectively (%).

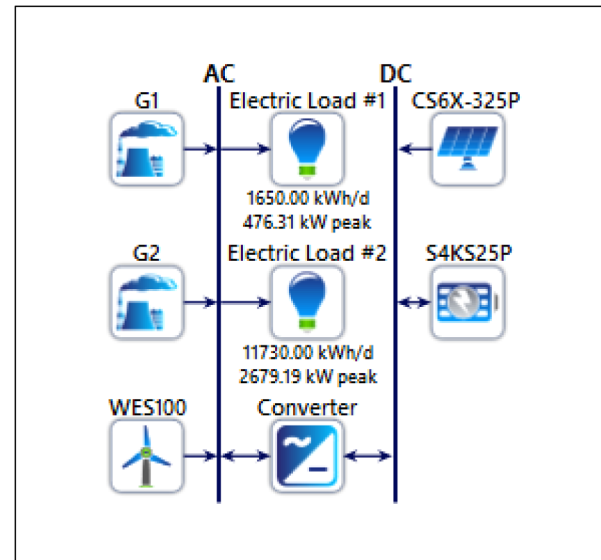


Fig. 5. Overall schematic diagram of the proposed HRES.

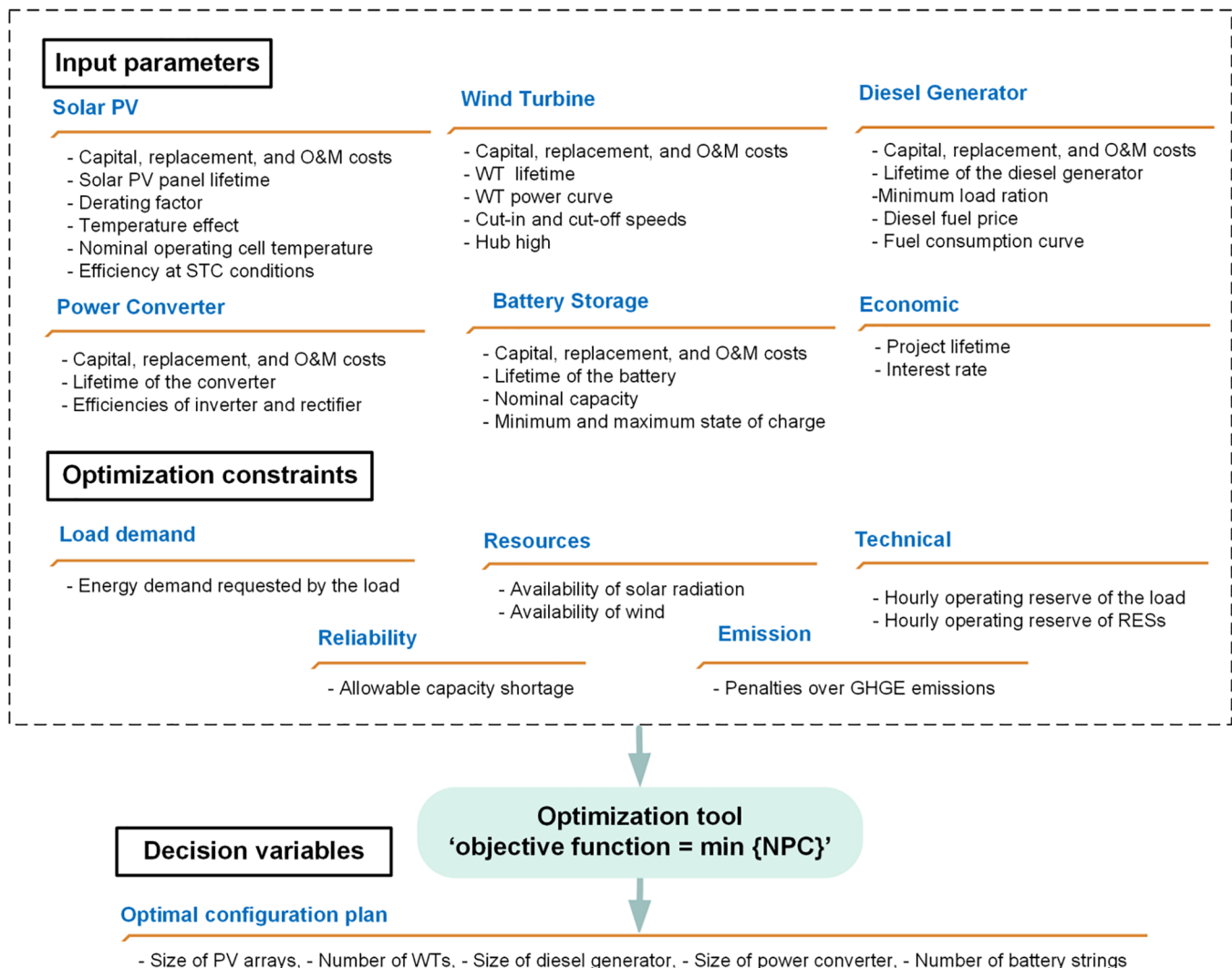


Fig. 4. Input parameters, decision variables, and constraints utilized for the optimization process of HRES planning and design.

Finally, α is the PV solar absorptance (%), while τ is the solar transmittance of any cover over the PV array (%).

2.2.1.2. Wind turbine. By using wind energy conversion system, electrical energy can be extracted from the wind through a suitable wind energy conversion system. In this study, WES100 high performance and reliable wind turbine model developed by Wind Energy Solution Company was selected. The turbine has a rated capacity of 100 kW and 24 m hub height, besides a lifetime of 20 years. WES100 has cut-in and cut-off speeds of 3 m/s and 24 m/s respectively. Capital and replacement costs of the utilized WT were 200,000 \$, while the O&M cost was assumed 150 \$/year [30]. The output power from the WT was calculated by determining the wind velocity at the hub high of the turbine using Eq. (4), then calculating the power output of WT under the standard air density condition at each wind speed. After that, the real density values were used to modify the computed outpower by Eq. (6) [57].

$$v_w = v_a \times [\ln(H_{hub}/H_{anem})/\ln(H_a/H_0)] \quad (4)$$

$$P_{W,stp}(v_w) = \begin{cases} 0 & v_w \leq v_{ci} \\ \frac{v_w - v_{ci}}{v_n - v_{ci}} \cdot P_{w,r} & v_{ci} < v_w \leq v_n \\ P_{w,r} & v_n < v_w \leq v_{co} \\ 0 & v_w \geq v_{co} \end{cases} \quad (5)$$

$$P_w = P_{W,stp} \times (\rho/\rho_0) \quad (6)$$

where, v_w and v_a are the wind speeds at the hub height (m/s) and anemometer height (m/s). H_{hub} , H_{anem} and H_a are the hub height (m), the surface roughness length (m) and the anemometer height (m), respectively. Also, P_w is the turbine output power (kW), $P_{w,stp}$ is the WT power output at the standard values of pressure and temperature, v_r , v_{ci} and v_{co} are rated, cut-in and cut-off speed of the selected WT, ρ is the actual density of the air (kg/m^3), $P_{w,r}$ is the WT rated power, and finally ρ_0 is density of the air at the standardized conditions (equals 1.225 kg/m^3).

2.2.1.3. Diesel generator. Two diesel generators have been used in this study. The first one was the existing generator in the investigated area and known as DG1 with a rated capacity of 500 kW; thus it has a zero-capital cost. The second generator denoted by DG2 was added as a part of energy system planning. Generally, the main purpose of DGs is to keep the utmost level of system reliability during the intermittent period of RE resources. The capital of DG2 was suggested to be 1000 \$/kW [21]. For DG1 and DG2, the replacement and O&M costs were assumed respectively as 900\$/kW and 0.39\$/hr. The minimum load ratio of the two generators was 25%. Besides, the lifetime of in hours was selected to be 24,000.0 hr and the diesel fuel price was considered as 1.28 \$/L [23]. The diesel fuel consumption basically depends on the power output of the DG, and it can be expressed as follows [21]:

$$F_{DG,C} = A_{DG} \times P_{DG,r} + B_{DG} \times P_{DG,g} \quad (7)$$

where, $F_{DG,C}$ is the fuel consumption rate (L/h), $P_{DG,r}$ and $P_{DG,g}$ are the rated power and supplied power of the DG (kW), while A (0.0140 L/kW/hr) and B (0.2440 L/kWh/hr) are the coefficients of the diesel fuel consumption curve.

2.2.1.4. Power converter. With the integration of DC power sources like PV and utilization of ESS, while the load demand is AC, it is essential to convert the DC electricity to AC and vice versa. So, a power converter is a vital component for this purpose. In this study, the capital and replacement costs per kW for the converter were estimated to be 515 \$/kW. The modeled converter's lifetime was 15 years with an efficiency of 95%.

2.2.1.5. Battery storage. A battery is one of the main components for achieving the balance between load and generation in HRES. It is used

mainly to offset the energy shortage of the available power sources or to cover the sudden increase in load demand. Thus, it assists in improving system reliability and stability. In this study, Surrette 4KS25P type was used and connected as a central storage system at DC bus and characterized by the capability to provide an enormous current for a short time interval, so it the most popular option in hybrid systems [62]. The capital and replacement costs of one battery were 1250 \$ and 1100 \$, respectively, while the O&M cost was assumed 15 \$/year [23]. The considered battery has 4 Vdc nominal voltage per string and 7.55 kWh nominal capacity. According to manufacture characteristics, the battery lifetime (or throughput) in kWh was 10,551.70, while the minimum state of charge (SOC) was 40%. Eq. (8) was used to calculate the storage capacity of the considered batteries [57]:

$$C_{wh} = E_L \times AD \times \eta_{Conv} \times \eta_{BAT} \times DOD \quad (8)$$

where, E_L is total energy requested by the load (kWh/day), the AD is daily autonomy; DOD is the depth of discharge of the battery, and η_{conv} and η_{bat} are the converter and battery efficiency, respectively

2.2.2. Representation of renewable energy resources uncertainty

To have a realistic modeling and making better decisions in HRES planning and design, the uncertainty of RE resources must be considered. Probabilistic Monte Carlo simulation method is the most common to describe and handle the power system uncertainties [14]. In this regards, the uncertainties related to solar radiation and wind speed were implemented by specifying them using probability distribution probability density functions (PDFs) [55], showing the range of values that a parameter could take, and also accounting for the probability of the occurrence of each value in the defined range [63]. In this paper, the random behavior of solar irradiance was characterized by Beta distribution function, as in Eq. (9), while the stochastic wind speed was modeled as a Weibull PDF which is defined by Eq. (12). The mathematical modeling of the two PDFs was represented according to [64]:

$$f_s(T_s) = \begin{cases} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \times T_s^{\alpha-1} \times (1-T_s)^{\beta-1} & \text{for } 0 \leq T_s \leq 1, \alpha \geq 0, \beta \geq 0, \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

where, $\Gamma()$ is the Gamma function, while α and β are the shape parameters of the Beta distribution of the solar irradiance, which can be determined using the mean (μ_{T_s}) and standard deviation (σ_{T_s}) of the solar irradiance calculated from measured data in a time interval t as follows:

$$\beta^t = (1 - \mu_{T_s}^t) \times \left(\frac{\mu_{T_s}^t \times (1 + \mu_{T_s}^t)}{(\sigma_{T_s}^t)^2} - 1 \right) \quad (10)$$

$$\alpha^t = \frac{\mu_{T_s}^t \times \beta^t}{1 - \mu_{T_s}^t} \quad (11)$$

$$f_{V_w}(v_w) = \frac{k}{C} \times \left(\frac{v_w}{C} \right)^{k-1} \times e^{-\left(\frac{v_w}{C} \right)^k} \quad (12)$$

The cumulative density function (CDF) for the Weibull PDF is given by Eq. (12), where its inverse is used to calculate the wind speed as in Eq. (13).

$$F_{V_w}(v_w) = 1 - e^{-\left(\frac{v_w}{C} \right)^k} \quad (13)$$

$$v_w = C \times (-\ln(r))^{\frac{1}{k}} \quad (14)$$

where, C and k denote the scale and shape factors of the Weibull function; r is a random number uniformly distributed on $[0, 1]$. Practically, the factors of Weibull function C and k can be calculated, approximately, using mean (μ_v^t) and standard deviation (σ_v^t) of the wind speed measurements in considered time interval t as follows:

$$k^t = \left(\frac{\sigma_v^t}{\mu_v^t} \right)^{-1.086} \quad (15)$$

$$C^t = \frac{\mu_v^t}{\Gamma(1+1/k^t)} \quad (16)$$

2.2.3. Economic model development

The total system NPC is defined as the present value of all the costs of installing and operating the system incurred over the project lifetime minus the present value of all the costs of installing and operating the system. The expression to calculate the value of the total NPC (\$), and the objective function (i.e. minimization of total system life-cycle cost) is given by [61]:

$$ObjF = \min(NPC) = \min(C_{tot,an}/CRF(d, T_{proj})) \quad (17)$$

where, $C_{tot, an}$ is the total system annualized cost (\$/yr) and $CRF(d, T_{proj})$ is the capital recovery factor (also, called discounting factor), d and T_{proj} are the project discount rate (interest rate) on capital investments of the installed equipment (%) and the project lifetime (yr), respectively.

The total system annualized cost is the summation of capital, replacement, O&M costs and salvage costs of system components and can be calculated using Eq. (18). While the discounting factor is given by Eq. (19).

$$C_{TOT,an} = \sum_{K \in PV, WT, DG1, DG2, Battery, Converter} C_{cap,k} + C_{O \& M,k} + C_{repl,k} - C_{salv,k} \quad (18)$$

$$CRF(d, T_{proj}) = d \times (1 + d)^{T_{proj}} / [(1 + d)^{T_{proj}} - 1] \quad (19)$$

where, denotes system component either PV, WT, DG1, DG2, battery, converter, $C_{cap,k}$, $C_{O \& M,k}$, $C_{repl,k}$ and $C_{salv,k}$ are respectively the total capital and installation cost, total operation and maintenance cost, total replacement cost, and total salvage cost of each system component and are expressed by Eqs. (21) and (22) respectively [65]. It should mention that a straight line or linear depreciation method of system components was applied in this study, meaning that the salvage value of a component is directly proportional to its remaining life [66]. Also, it was assumed that the salvage value depends on the replacement cost rather than the initial capital cost.

$$C_{cap,k} = N_k \times CU_{cap,k} \quad (20)$$

$$C_{O \& M,k} = N_k \times CU_{O \& M,k} \quad (21)$$

$$C_{repl,k} = N_k \times CU_{repl,k} \times SFF(d, T_k) \times F_{repl,k} \quad (22)$$

$$C_{salv,k} = C_{repl,k} \times T_{rem,k} / T_k \quad (23)$$

$$T_{rem,k} = T_k - (T_{proj} - T_{repl,k}) \quad (24)$$

where, N_k refers to the total number of components from the same type, $CU_{cap,k}$, $CU_{O \& M,k}$, $CU_{repl,k}$ and $CU_{salv,k}$ are respectively the component capital and installation unit cost, annualized unit cost for operation and maintenance, annualized replacement unit cost, and salvage unit cost (\$), $T_{rem,k}$, T_k and $T_{repl,k}$ and are the remaining life of the component at the end of the project lifetime (yr), the component lifetime (yr) and the corresponding replacement cost duration (yr), $SFF(d, T_k)$ is the funding factor, $F_{repl,k}$ is a factor arising when the component lifetime differs from project lifetime. $SFF(d, T_k)$ and $F_{repl,k}$ can be calculated from the following equations:

$$SFF(d, T_k) = d / [(1 + d)^{T_{proj}} - 1] \quad (25)$$

$$F_{repl,k} = \begin{cases} CRF(d, T_k) / CRF(d, T_{proj}) & T_{repl} > 0 \\ 0 & T_{repl} = 0 \end{cases} \quad (26)$$

The system with a minimum life-cycle cost ultimately results in a

system with a minimum levelized cost of energy. Therefore, COE (\$/kWh) is considered as a valuable investigator when assessing the economic characteristics of the HRES. COE is defined as the average cost per kWh of the useful electrical energy produced by the system and it is expressed as in Eq. (27) [61]. Additionally, the return on investment (ROI) is another important criteria, which used to analyze the economic viability or the profitability of the HRES as an investment [60]. ROI is defined as the yearly cost saving relative to the initial capital investment on the reference system and calculated by Eq. (28) [67]. In this study, worth to declare that the reference system is referred to as the base case using the diesel generator only.

$$COE = C_{tot,an} / E_{load} \quad (27)$$

$$ROI = \sum_{i=1}^{T_{proj}} (C_{i,ref} - C_{i,curr}) / [T_{proj} \times (C_{cap,curr} - C_{cap,ref})] \quad (28)$$

where, E_{load} is the total electrical load served (kWh/yr) by HRES (kWh/yr), $C_{i,ref}$ is the nominal annual cash flow of the reference (base) system, $C_{i,curr}$ is the nominal annual cash flow of the current system, $C_{cap, curr}$ is the capital cost of the of the reference system, and $C_{cap, ref}$ is the capital cost of the reference system. In the meantime, the electrical energy generated from RESs and delivered to the load is expressed by renewable fraction (RF) of the hybrid system and calculated as follows:

$$RF = 1 - (E_{g,non-ren} / E_{load}) \quad (29)$$

where, $E_{g,non-ren}$ is the total electrical energy produced from CESs (i.e. non-RESs) (kWh/yr).

2.2.4. Environmental emission model development

The presence of conventional power sources would lead to producing a certain amount of carbon gases, mainly CO₂. The main environmental concern addressed in this study is to calculate the total harmful CO₂ emissions associated with system components. The amount of CO₂ emitted from the hybrid system is related directly to the amount of fuel consumption. This value is calculated from the following equation [68]:

$$TCO_2 = 3.667 \times m_f \times LHV_f \times CEF_f \times X_c \quad (30)$$

where, TCO_2 is the total amount of emitted CO₂, m_f is the amount of diesel fuel (L), LHV_f is the fuel heating value (MJ/L), CEF_f is a factor related to carbon emission (ton carbon/TJ), and finally X_c is the oxidized carbon fraction, where each 3.667 gm of CO₂ includes a quantity of 1 gm of carbon.

2.2.5. Study scenarios

In grid-isolated systems, the utilization of the diesel generator systems usually is the first choice for supplying the load demand, due to their low capital cost compared to RE technologies. However, the adverse implications of diesel systems (such as harmful GHGE, noise pollution, expensive values of O&M and fuel costs) implies more limitations on their utilization. On the other side, RESs are clean, relatively free O&M cost with inexhaustible resources. However, RE resources have a periodic nature. Thus their output capacity is not reliable and not precisely predictable. This problem can be solved by energy storage, but it is expected to be economically infeasible. Considering that, mixing of different energy sources as a hybrid system may solve previous problems related to technical, economic and environmental aspects. In this paper, to cover all possible combinations of the available power sources, five different cases were modeled and analyzed to identify the most satisfying configuration of the HRES. The studied scenarios were: Case 1: PV-DG1-DG2-Battery-Converter, Case 2: WT-DG1-DG2-Battery-Converter, Case 3: PV-WT-DG1-DG2-Battery-Converter, Case 4: PV-WT-DG1-Battery-Converter, and Case 5: DG1-DG2. The latter case was referred to as the base case for the sake of comparison in this study.

2.3. HOMER Pro® optimization tool

Several optimization tools reviewed in [69], have been used in literature for optimal planning and design of HRESs including HOMER Pro®, RETScreen [70], Hybrid2 [71] and iHOGA [72]. Among them, HOMER Pro®, or HOMER (Hybrid Optimization of Multiple Electric Renewables) is founded as the most integrated and powerful tool that is widely recommended and used in countless studies in the literature. Recently, HOMER has been recommended and utilized by several researches as the global standard for optimal planning and design of HRES in all energy sectors. For example, radio telecommunication station [45], desalination plants [47], rural households [48], urban cities [57], large residential communities [63], coastal regions [69]. Moreover, the authors in [15] introduced a comprehensive survey of the studies that applied HOMER exclusively for planning and design optimization of HRESs. So, HOMER Pro® was chosen in this work to formulate and model the design optimization study. HOMER was developed by National Renewable Energy Laboratory, USA for techno-economic modeling, simulation, and optimization of HRESs [73]. It can help system modelers and designers to build an accurate economic model and check the technical specification, sizing and boundaries for different types of energy technologies in order to evaluate the practicality of the optimal HRE configuration out of many possibilities before the system is installed and commissioned [74].

In this regards, the developed method steps required to study the feasibility of the HRES and to find the optimal configuration plan is presented in Fig. 6. Firstly, a pre-HOMER analysis was performed by assessing the load profile and the availability of RE resources according to the proposed design framework in Fig. 3. Then, by gathering and handling all input parameters and design constraints mentioned above in Fig. 4 such as load demand profile beside solar and wind resources data of the studied area, as well as the design and economic parameters of system components. After that, all data were defined to HOMER Pro® interface. Then, all possible scenarios of the hybridization of the adopted energy technologies were explored. Next, the simulation and optimization analysis were conducted for each scenario based on an hourly basis and subjected to system constraints [27]. Only feasible solutions which ensure load balance between generation and consumption will be considered in the optimization process [16]. So, for each feasible solution, system optimization was performed, in which a proprietary derivative-free algorithm is used to search for the least-costly system configuration [30]. Once NPC precision convergence criterion is met achieved, the optimization is completed. Finally, after optimizing the all feasible solutions, the list of optimal system configurations plans was provided was obtained and ranked based on the total NPC. Among all optimized configurations, only one case has been nominated as the optimal solution (or winning configuration plan) with the lowest NPC.

2.4. Operating dispatch strategy

The selection of the proper energy dispatch strategy that controls system operation and manages the flows of energy between different system components (i.e., determine which equipment must operate at the time and its reference power) is essential. Two types of strategies are provided by HOMER Pro, called; cycle charging (CC) and load following (LF) [61]. In the first one, the system diesel generator operates at its full capacity to primarily serve the load demand as well as the required operating reserve. While the surplus power is used to serve the deferrable load and electrolyzer, if any, as well as to charge the battery storage until it reaches the set point state of charge. In the second strategy, the main priority of the system diesel generator is to generate only a sufficient power to fulfill load demand at the time, and it is not required to give other surplus power for serving other objectives. These tasks are the responsibility of renewables. In this sense, the implemented dispatch strategy for the considered hybrid energy system

was based on the load following control concept. This strategy tends to be suitable for systems with a high penetration of renewables [49]. Also, LF strategy is expected to minimize the total NPC than CC strategy. This because the former will reduce the operating hours of the diesel generator, and consequently the amount of consumed fuel. Moreover, preventing the battery from the frequent charge/discharge process will increase its lifetime.

Fig. 7 demonstrate how the flow of energy is managed by LF strategy at each time step of the simulation. From the given flow chart, the first operation mode is when the calculated output power from PV and WT equal to the required load demand and operating reserve, which mean that there is no any excess power. In that case, the PV and WT will be used mainly to serve load requirements and the battery will not absorb any power and kept at its current state of charge at the time. Meanwhile, the diesel generators will stay in a shut-down state. In the second operation mode, if the output power from renewables is larger than the load demand. It will be used to fulfill load requirement and then the excess power will be used to charge the battery as long as the battery is not fully charged (SOC less than 100%). If not, the excess electricity could be used as a dump load for water or space heating. In this second operation mode, the diesel generators also will remain off. The last operation mode is activated when the output power from PV and WT is less than the load demand (i.e. there is a certain unmet load). Here there are two possible states; first, if SOC of system battery is equal to its minimum value, the diesel generators will turn on to only provide a sufficient power to serve the unmet load and without feeding the battery. Also, any excess power from the PV or WT will be used again to charge the battery according to the SOC of the battery. The second state is when SOC of the battery is greater than the minimum SOC. In such a case, the battery will be discharged to serve the unmet load until the load satisfied. If not, the diesel generators will operate to feed the load and cover the power shortage in renewables and batteries.

3. Case study

To demonstrate the proposed framework for HRES planning design optimization, the data of specific agriculture and irrigation area in Dongola, Sudan were acquired and presented. First, the potential of RE with the current status of the energy sector in Sudan is introduced. After that, the data of meteorological features of the investigated area together with the estimated electrical profile described fully in the upcoming subsections.

3.1. Renewable energy prospects in Sudan

As a part of Arab countries and located at the Middle East and North Africa (MENA) region, Sudan has a high potential of PV and WT energy systems projects deployment, especially in North area [75]. This due to the availability of renewable energy of resources (i.e. wind and solar) over the year. Fig. 8 shows Sudan's solar atlas [76] and wind atlases [77] obtained from the World Bank Group. The two maps demonstrate the distribution of average solar radiation and average wind speed over Sudan, whereas the average values of solar radiation and wind speed recorded around 6.5 kWh/m²/day and 6.0 m/s, respectively, thus they are measured as among the highest values in the world.

As reported by IRENA, in 2015, only 0.38% (12 MW with excluding hydro) of the total Soudan's installed capacity was counted by renewables [78], while the remaining installed capacity was divided between hydro and conventional generation systems. With increasing environmental awareness and moving toward more sustainable energy systems, in 2014 IRENA was partnered with the League of Arab States (LAS) the regional center for Renewable Energy and Energy Efficiency (RCREEE) to identify as a set of actions to encourage and increase the investments in renewables energy projects in Arab region, thus Pan-Arab renewable energy strategy 2030 has been established [78]. As a result for this strategy, the Sudan government plan targets to increase the total

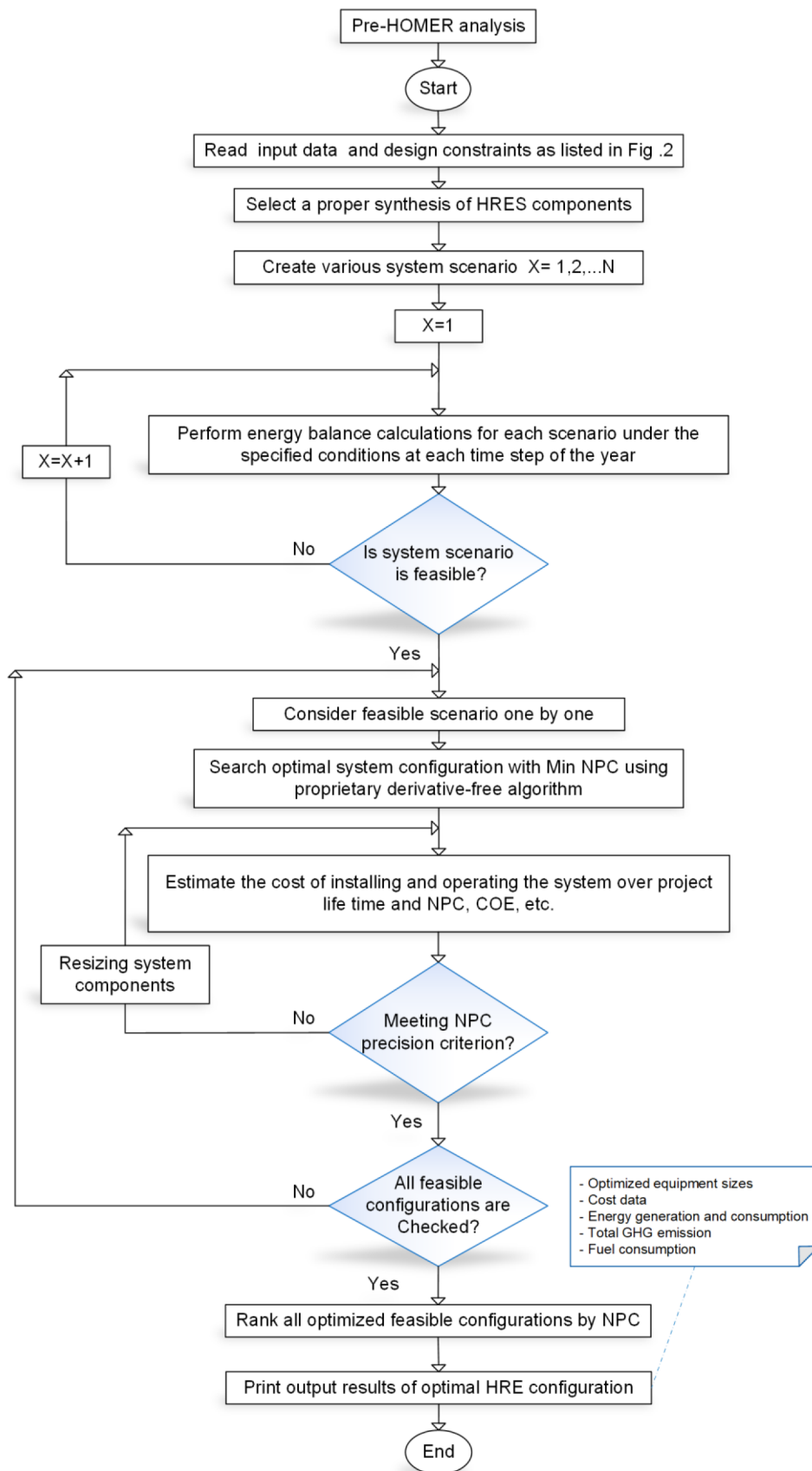


Fig. 6. Method used for techno-economic simulation and optimization.

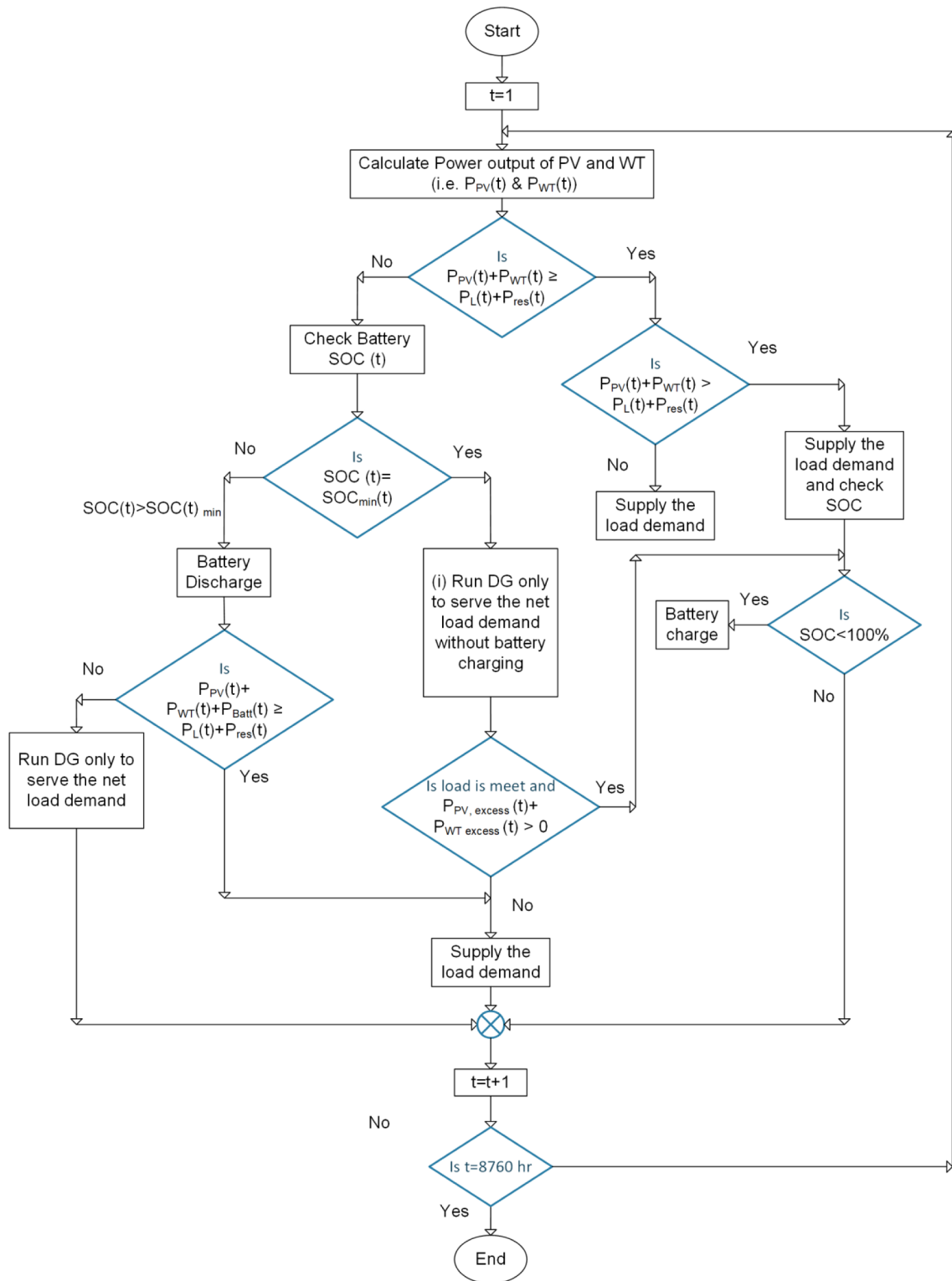


Fig. 7. Energy dispatch strategy for the developed PV-WT-DG-battery-converter HRES.

installed capacity of renewables to 11% by 2020 and considering a larger sharing from renewables to reach up to 20% (1000 MW PV, 1000 MW WT, 100 MW concentrating solar cell) by 2030.

Notwithstanding the great efforts made by local utilities in Sudan to address the electricity sector's bottlenecks, only 46% of the population in Sudan have a reliable and safe source of electrical energy according to International Energy Agency statistic in 2016 [79]. While the remaining (~22 million people) are totally deprived of electricity or fed

by a standalone diesel generator. Besides, Sudan is considered one of the least developed countries, where it's based mainly on the agricultural sector, and it is badly affected by electricity shortage. Also, to the best of author's knowledge, there is no work has been done in the literature with a strategic context to study specifically the feasibility investigation of HRES in Sudan despite the abundance of solar and wind resources. So, it is expected that this study will assist in achieving the target of Sudan's government toward clean and reliable energy supply

for all, and also highlight the potentiality awareness of RE in Sudan from a practical point of view.

3.2. Site description and renewable resources data for Dongola, Sudan

The selected site of the studied area is located geographically at 19° 10' N latitude and 30° 28' E longitude in Dongola city in North state (see Fig. 8). The accurate climate parameters of the selected site were acquired from NASA surface metrology and solar energy database [80]. Where, the annual average solar radiation and annual average wind speed were founded to be 6.28 kWh/m²/day and 4.21 m/s, respectively. Also, the annual average ambient air temperature was 27.8 °C. Fig. 9 shows the monthly averaged global solar radiation and the ambient air temperature profiles over a 22 year period (July 1983–June 2005) and the monthly averaged wind speed profile over a 10 year period (July 1983–June 1993). The provided graphs illustrate the variation tendency of solar radiation and wind speed potential over the whole year. Apparently, the average solar radiation increases wavyly toward peak value in June, then start to go gradually downward turn to the lowest value in December. Also, it was seen that the mean wind speed at the beginning and end of the year is enormously higher than summer time.

Fig. 10 shows the solar radiation and wind speed PDFs derived from the Beta and Weibull distribution, respectively. It is observed that the extreme occurrence frequencies are, approximately, skewed near the higher values of average solar irradiance and wind speed; the peaks further indicates the most frequent values. Further, the cumulative PDF of solar radiation and wind speed, as shown in Fig. 11. For solar irradiance greater than 0.25 kW/m² and wind speeds greater or equal to 3.0 m/s cut-in wind speed, the investigated site has frequencies more than 60% and 35% for solar radiation and wind speed. This is very encouraging outcome for the installation of RES on the site of Dongola.

3.3. Electrical load demand estimation

The studied area was located near the Nile River as a suitable site for agriculture reclamation and irrigation project of around 1000 Hectare (10,000 m²) in Dongola. For this purpose, the estimated quantity of required water per day was given to be 40,000 m³ per day. The first phase of the project has been already completed with a successful reclamation of around 400 Hectare, where the existing electrical energy system is isolated from the national grid of Sudan and consisted from one standalone diesel generator, which is denoted by DG1 in this study. The rated power of the utilized generator was 500 kW and was used to provide the electrical power for two water irrigation pumps. Each pump has a rated power of 165 kW and water capacity of 1000 m³ per hour as well as a head high of 50 m. The purpose of the two pumps was to lift the water from the Nile River to irrigate the existing agricultural area of 400 Hectare. The water pumps were used mostly during five daylight hours from 10:00 AM to 2:00 PM, which maximize the summer peak load demand over the daytimes. The second phase of the project targets to finish the irrigation of the remaining 600 Hectare. So, another three pumps were added by system designers to the existing two pumps with the same specifications. Moreover, the electrical load list of the investigated area was expanded to include other categories of loads for domestic and commercial applications. The additional electrical loads that were established plus the irrigation water pumps include water treatment plant, bottle plant, stores and fridges for agricultural crops, sales shops, and employee offices. Consequently, this increases the total peak power of the second load to reach around 2679 kW. Aggregated electrical load list and the corresponding daily operating hours, as well as the utilization hours of each load, are provided in Table 2. Daily load profile showing the load demand variance throughout 24 hr of the day illustrated in Fig. 12a, where the average energy consumption of both load 1 and load 2 are 1650 kWh/day/year and 11,730 kWh/day/year, respectively. To make the load data more realistic and practical, the

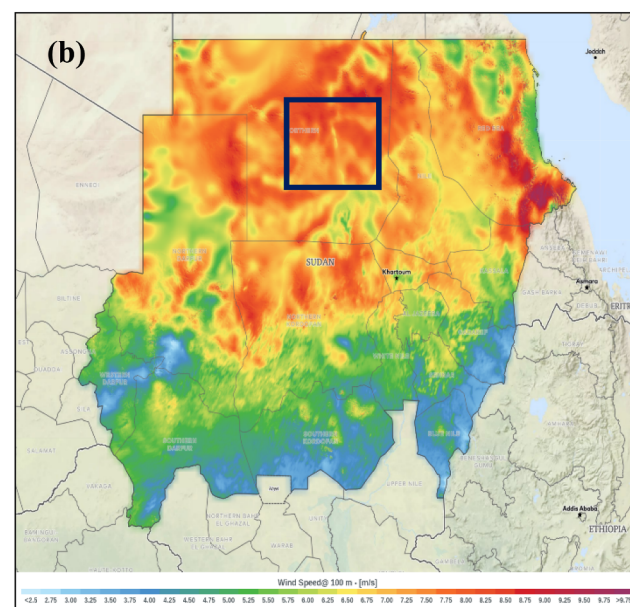
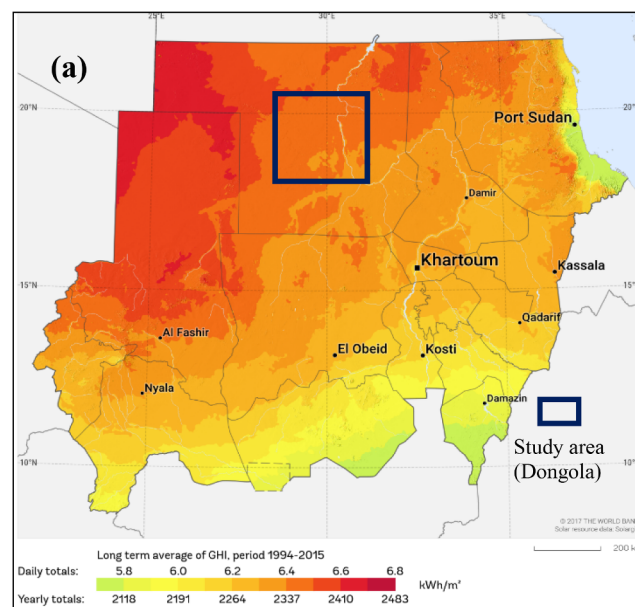


Fig. 8. Global atlases for solar and wind resources, Sudan: (a) Sudan's solar energy potential [76], (b) Sudan's wind energy potential [77].

data were synthesized by specifying typical daily electrical loads profiles, and then the randomness was added. In this sense, random variability of 10% day-to-day and 10% time-step-to-time-step for the electrical load demand was adopted in this study. The monthly electrical load profile is also presented in Fig. 12b.

3.4. Study constraints

Generally, in many optimization problems, certain functional constraints and requirements must be satisfied through the design variables. In this study, the optimization was performed under a number of constraints and assumptions that can be outlined as follows:

- Resources constraints: as mentioned above, the available data of RE resources at the local level in Dongola are: solar resource with

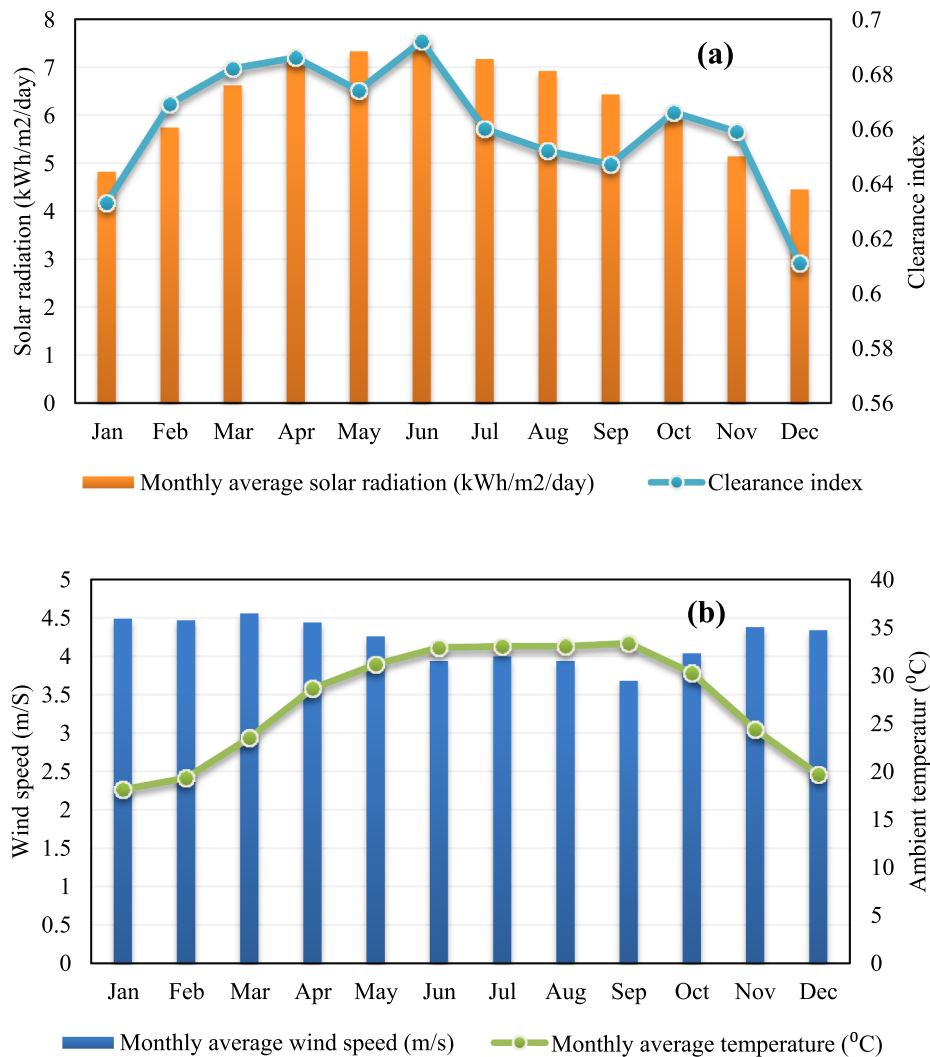


Fig. 9. Renewable energy resources for Dongola [80]: (a) Monthly average solar radiation profile and (b), Monthly average wind speed and ambient temperature profile.

6.28 kWh/m²/day (yearly average solar radiation) and wind resource with 4.21 m/s (yearly average wind speed);

- **Economic:** the project lifetime was considered as twenty-five (25) years because the governments usually attempt to extend the harnessing of RE for a very long time along. Also, the annual real discount rate was presumed to be 6% as a typical commercial value. On the other hand, there is no fixed initial cost or operating costs were considered in this study in order to provide convenience in the economic comparison;
- **Technical:** maintaining power balance between generation and consumption as well as keeping the SOC of the battery within the safe permissible range are the primary technical constraints adopted in this study. Besides, the hourly load operating reserve was set to be 5% for accounting the sudden spikes in the load, while a 5% operating reserve was considered for the PV and WT output power due to the inherent dependency of the two sources on weather conditions;
- **Reliability:** the maximum capacity shortage (or loss of power supply probability index) was suggested to be zero to ensure a feasible solution with a high level of power supply reliability; and
- **Emission:** in the current situation, the CO₂ emission trade scheme is not available in Sudan. So, there is no penalty over CO₂ emissions has been taken into account in this study.

4. Results and discussion

This section includes the results of various studies performed in the current work. A discussion of each of these results is also highlighted.

4.1. Techno-economic optimized results of the developed hybrid renewable energy system for Dongola

By integrating different input parameters with the involved objective function and the assumption adopted in this work, the techno-economic analysis and evaluation of the various hybrid system configurations were performed. Long-term optimized simulation results considering the stochastic behavior of RE resources are summarized in Table 3 and Table 4. Where the summary of economic and environmental investigators is introduced in Table 3 including NPC, COE, ROI and the corresponding reduction of NPC and COE for each case compared to the base case at the last column. Also, the same table provides the value of the total system GHG emission, fuel consumption, annual operating hours (OH) of diesel generators and the percentage of RE share is also provided and denoted by RF as well. On the same line, the optimal capacities of each individual component for each configuration and the summary of different cost items are given in Table 4. In general, the criteria of power balance between electricity generation and demand were achieved for each configuration with the provided

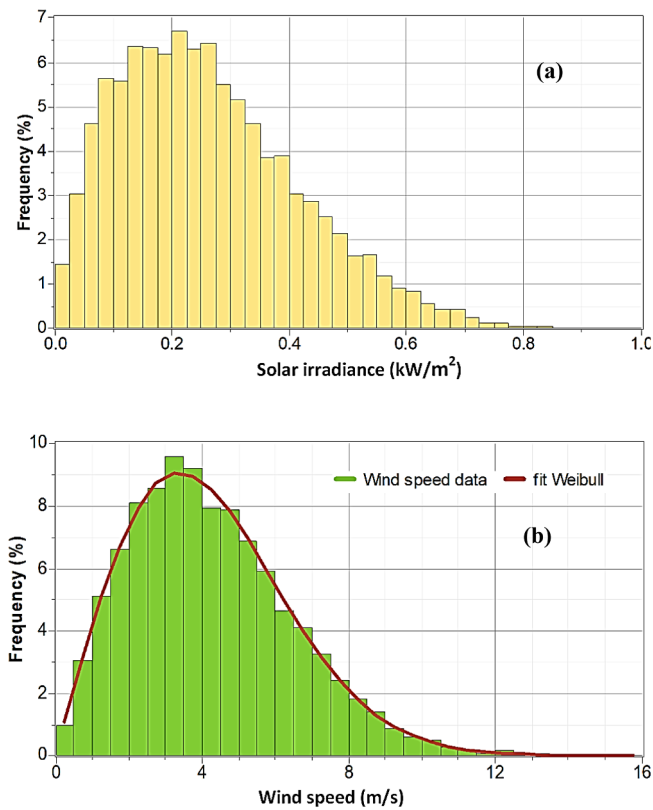


Fig. 10. PDFs of RE resources for Dongola (a) Beta distribution of solar irradiance, (b) Weibull distribution of wind speed.

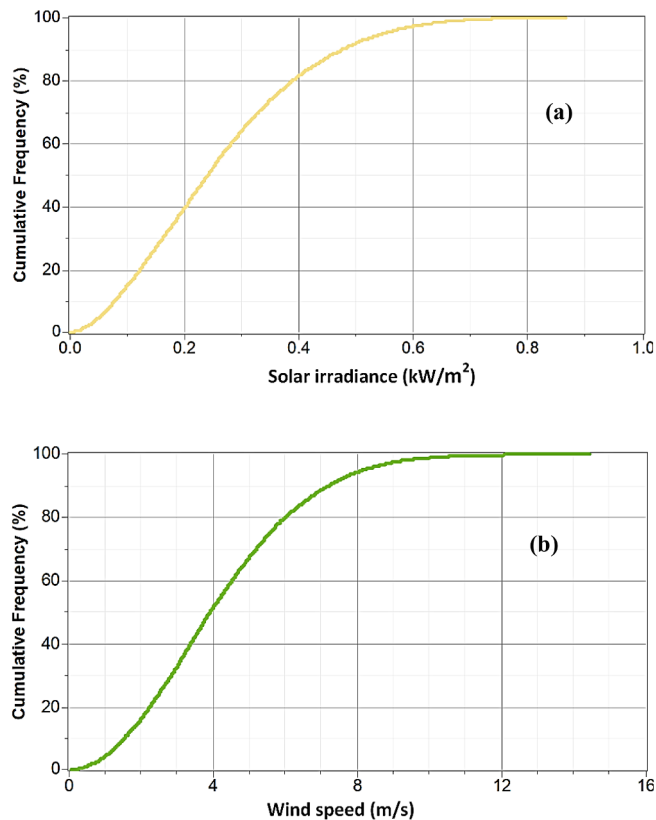


Fig. 11. (a) Cumulative density functions of RE resources for Dongola (a) Cumulative frequency of solar irradiance, (b) Cumulative frequency of wind speed.

Table 2

Electrical load list established for agriculture and irrigation area in Dongola.

Load Type	Power (kW peak)	Working hours/day	Utilization time	Total kWh/day
Load 1(1st phase)				1165
Water irrigation pump	2 × 165	5	10:00–15:00	
Load 2 (2nd phase)				11,730
Water pump	3 × 165	5	10:00–15:00	
Bottle Plant (capacity: 5000 bottles with 1 L, each)	1000	5	10:00–15:00	
Water treatment plant	56	5	10:00–15:00	
Fridges	56	24	0:00–24:00	
Sales outlet	12	16	0:00–9:00 & 17:00–24:00	
	56	8	9:00–17:00	
Offices & Stores	45	16	0:00–9:00 & 17:00–24:00	
	150	8	9:00–17:00	

equipment ratings. Fig. 13 shows the optimization plot of the total NPC against CO₂ emissions with zoomed screen of the least-costly feasible solutions. Actually, there 9568 solutions or system configurations were simulated, among them there are 6005 were feasible solutions (each simulation case is symbolized as a single point), while the remaining solutions were omitted due to optimization constraints. Each viable configuration was optimized to get the best plan. Among all these optimized configurations, only one case has been nominated as the optimal solution (i.e. winning configuration plan) with the lowest dot as displayed by zooming into the areas of interest as pictured below in Fig. 13b. Further, the optimized components' sizes are mentioned, which are the same as in Table 4. For comparative purpose, the variation tendency of economic and environmental metrics of different cases are graphically presented in Fig. 14.

The main points that are highlighted from the obtained optimized results for the HRES are summarized as follows:

- Among all studied cases, the combination of PV-WT-DG1-DG2 with battery and system converter (case 3) shows optimal economic performance for electrification of the investigated area. So, this configuration was selected as the winning energy system to supply the investigated area. The system was recorded the minimum values for NPC and COE of 24.16 M\$ and 0.387 \$/kWh, respectively.
- According to the optimized sizing results, the proposed optimal HRES was comprised of 2000 kW PV arrays, 42 WTs (100 kW each), 1500 kW DG1 together with the existing DG2 of 500 kW rated power. Besides the optimal number of batteries was founded to be 3551 (7.55 kWh each) and converter rating was 2907 kW.
- The optimal configuration (or winning plan) was obtained with a high penetration of PV and WT, where the RF share was 95.12% of the total electricity production. Obviously, the optimal size of the PV arrays was the maximum available value (2000 kW) in all feasible plans given in Table 3. Therefore, it is expected that the PV will be the exceptionally the dominant RESs in the hybrid system once the topographical constraints of the investigated area are removed.
- The proposed optimal HRES was more costlier than the base case of the diesel system. Where significant reductions by about 76.89% in NPC and 76.90% in COE were achieved when the optimized hybrid system was used instead of the base case using DGs only. The optimal system also gets a positive value of ROI (46.1%), which mean it was received a high value of financial returns. Comparison of nominal cash flow results of the optimal case and the base case is presented in Fig. 15, indicating the optimal HRES has great benefits compared to the base case in saving and recovering expenditure. Undoubtedly, this clearly shows the economic feasibility of the proposed HRES over the 25-year project lifetime.

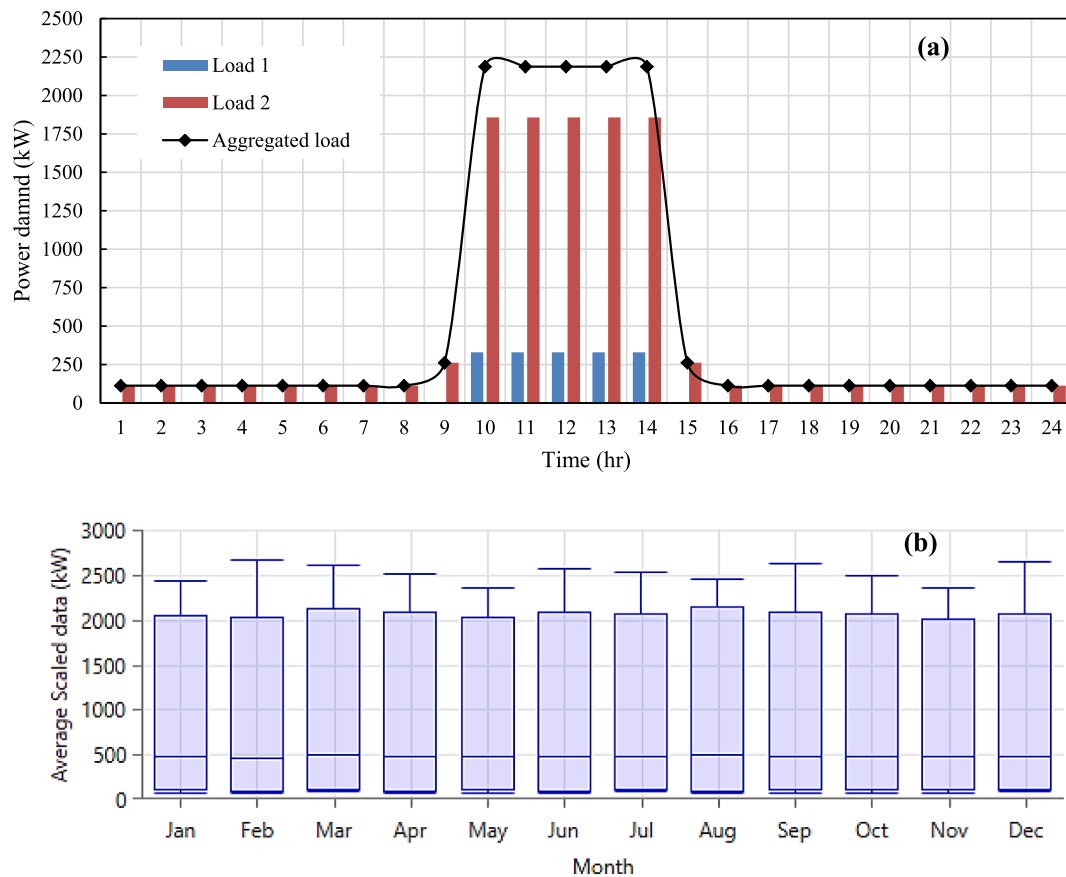


Fig. 12. Electrical load profile of the agriculture and irrigation area in Dongola: (a) Daily electrical load profile, (b) Seasonal electrical load profile.

Table 3

Summary of optimized results of different HRES configurations ranked by NPC.

Rank	Case	System configuration	NPC (M\$)	COE (\$/kWh)	ROI (%)	RF (%)	GHG (kg/yr)	Fuel (L/yr)	OH of DGs (hr/yr)		NPC&COE reduction (%)
									DG1	DG2	
1	3	PV-WT-DG1-DG2-Battery/Converter	24.16	0.387	46.1	95.12	168,786.3	63,798.2	120	229	77
2	4	PV-WT-DG1-Battery/Converter	28.38	0.455	30.9	99.84	5,599.566	2,116.7	34	–	73
3	1	PV-DG1-DG2-Battery/Converter	39.94	0.640	153.1	68.26	110,165.5.3	416,404.3	605	1621	62
4	2	WT-DG1-DG2-Battery/Converter	59.79	0.958	13.1	84.37	540,560.7	204,321.6	645	386	43
5	5	DG1-DG2	104.55	1.675	0	0.00	351,3972	132,8214	7398	1825	0

Table 4

Optimized equipment sizes and cost results associated with each study case.

Case	Equipment size						Costs (M\$)				
	PV (kW)	WT (Qty.)	DG1 (kW)	DG2 (kW)	Battery (Qty.)	Converter (kW)	Capital	Replacement	O&M	Fuel	Salvage
1	2000	–	500	1500	1752	1475	5.850	13.958	13.988	6.813	–0.260
2	–	134	500	2500	7091	3057	39.738	16.741	6.676	3.343	–6.711
3	2000	42	500	1500	3551	2907	17.236	6.180	2.470	1.044	–2.775
4	2000	63	500	–	7094	2889	24.355	7.419	0.563	0.035	–3.993
5	–	–	500	2500	–	–	2.500	39.115	41.188	21.733	0.000

- However, the fourth case using PV-WT-DG1-battery-converter system offers the most sustainable and environmental option (i.e. the lowest value of the total emitted gases of 5,599.57 kg/yr) with the higher share of RE of 99.84%, it was ranked as the second among all feasible solutions for Dongola from an economic point of view. This because such system was involved a high cost for NPC (28.38 M\$) and COE (0.455 \$/kWh), which was attributed to the

huge values of the initial capital as well as WT and battery replacement cost.

- Because of the lower value of the initial capital cost of the first case (PV-DG1-DG2-battery-converter), the system was received the most prominent value the financial return (153% ROI). But, the system was ranked the third among the whole studied cases from the economic point of view. This explained the bad effect of the remarkably

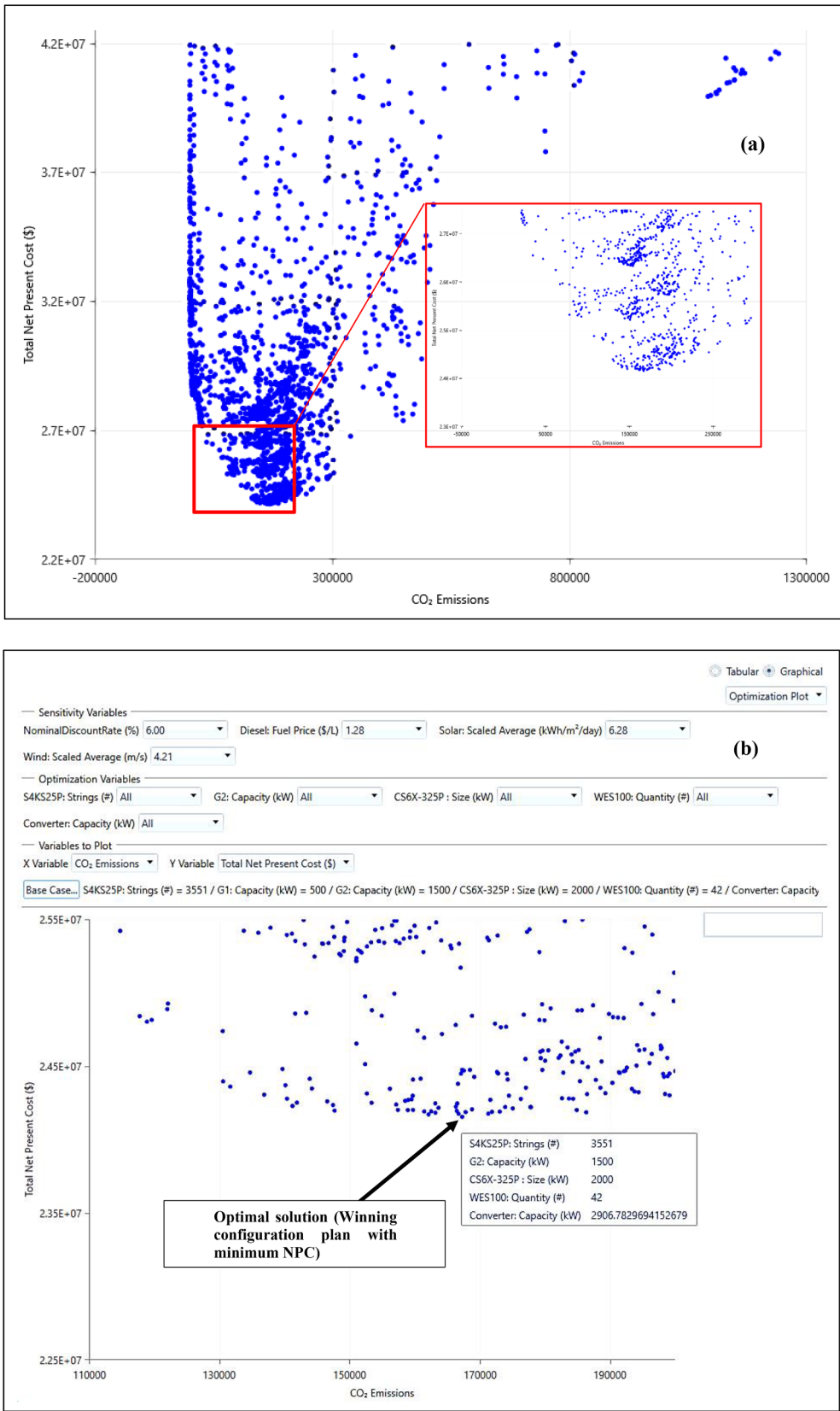


Fig. 13. (a) Optimization plot of the NPC against CO₂ emissions, (b) Zoom-in optimization plot on the area of interest with minimum NPC.

- high-value of replacement, O&M and fuel costs of diesel generators, particularly DG2.
- The analysis of the obtained results also shows that the base case which based on 100% conventional energy sources, has the lower value of the initial cost. This because the initial capital cost of the whole system (2.5 M\$) was accounted only by the capital and installation cost of DG2, together with the already existed diesel ‘DG1’ with zero capital cost. However, the O&M and replacement costs of

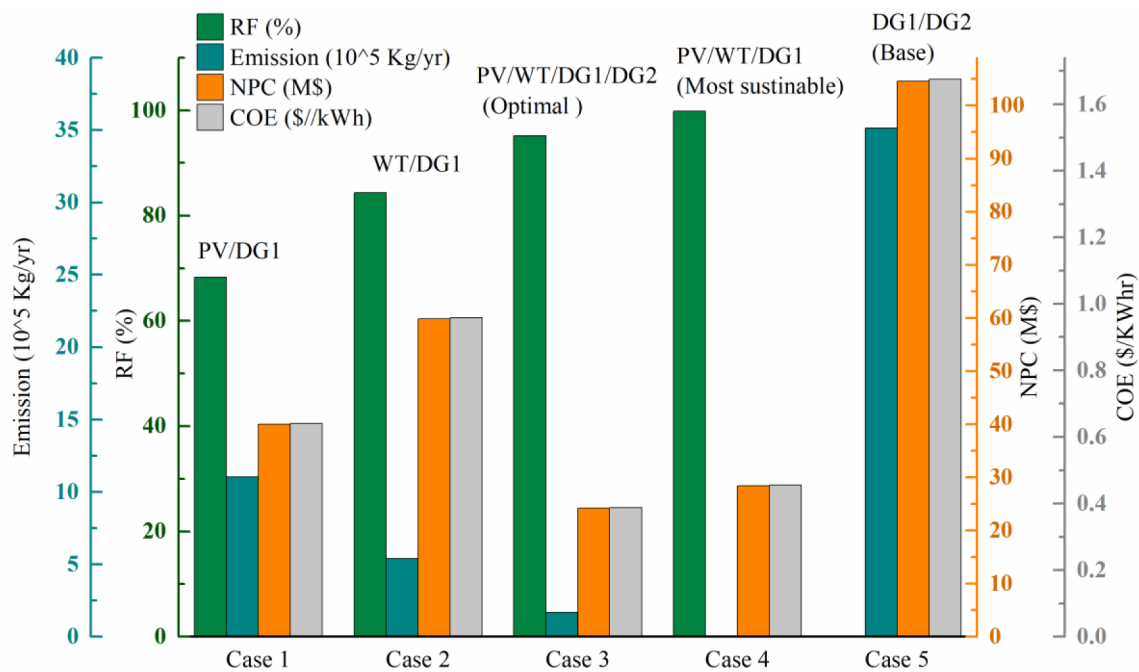


Fig. 14. Economic and environmental performance of the developed HRES configurations.

the overall system was founded to be larger than other cases with 41.188 M\$ and 39.115 M\$, respectively. By this means, the NPC has recorded the exceptionally maximum value among all cases (104.55 M\$), simultaneously with maximum COE of 1.675 \$/kWh. Besides, the system produces a significant amount of harmful GHG emissions equals to 351, 3972 kg/yr. Therefore, this system was regarded as the worst case from economic and environmental aspects.

4.2. Performance analysis and evaluation of the proposed optimal system configuration plan

From the previous discussion, it was concluded that PV-WT-DG1-DG2-Battery-Converter HRES gives a satisfactory result regarding technical, economic and environmental aspects which make it the optimal configuration plan for Dongola. So, the performance metrics of this system need to be analyzed.

4.2.1. Technical and operational performance

Technical performance assessment is one of the key parameters while optimizing the design of the HRES. The monthly average electric power production from each system component of the optimal configuration plan is displayed and compared in Fig. 16. Expectedly, the PV and WT were considered as the main sources to supply the aggregated electrical demand during the whole year due to the rich solar and wind in Dongola. In contrast, diesel generators were contributed by lower values in the annual electricity production. The later conventional systems were considered as a backup power source in most of the months when the output power from PV and WT are affected by weather conditions. For example; in August and September, the sharing of DG1 and DG2 was reached to its maximum values as a result of decreasing the average value of wind speed, and therefore the generated power from WT. Table 5 demonstrates the repartition of the total annual energy production of the HRES between the various power sources. From that, PV has contributed the higher value of energy generation by 3,820,438 kWh/yr (63.1%), while WT shared by 1,997,016 kWh/yr (33%) in total annual electricity production. This was attributed to the cheaper cost of the PV array in addition to the high availability of solar and wind resources in Dongola. In contrary,

there a non-substantial percentage of annual energy generation was evidently observed to be contributed by both G1 and G2. Where DG1 and DG2 share by just 3.933% of the total annual energy production. This reflects the effectiveness of the utilized LF energy dispatch strategy to decline the operating hours of diesel generators, thereby the fuel consumption rate as well as limiting the harmful GHG emissions. As a consequence, it is concluded that the optimal system trends to be independent of diesel generators in most of the loading conditions. Fig. 17 demonstrate the scheduling for electrical power generation and consumption over a one-day sample. It is seen from this figure that the optimal configuration plan was succeeded to meet 24-hr load demand with zero capacity shortage, as well as annual excess electricity generation of 584,545 kWh/yr, which can be used if a dump load such as heating or cooling demand will be available. Furthermore, the optimal hybrid system has an annual energy deficit of just 677 kWh/yr (0.0139% of the total power demand).

4.2.2. Economic performance

From an economic perspective, the cost parameters of the optimal hybrid system including capital, replacement, O&M, fuel, and salvage costs for system components and the overall system is summarized in Table 6. It was observed that due to the expensive capital cost of WT, it contributes the higher proportion of the total cost between system components with 8.40 M\$, which approximately equals 39.9% of the total system cost. But, it is expected that with the increasing mass production of wind energy system technologies, the WT capital cost will fall quickly in the near future [2], and this will lead to more reduction in NPC and COE. On the same line, both diesel generators also share by a considerable amount in the total system cost with 5.594 M\$, which predominantly attributed to increasing O&M and fuel costs of diesel generators. System's batteries and converter come in the second and third stage 4.974 M\$ and 2.005 M\$, respectively. Compared to all system components, PV was founded as the most affordable technology in the proposed HRES. This resulted from the nearly free O&M cost and low capital and installation cost PV based the speed development in material science and increasing the global trend in PV investments [11]. Therefore, this highlights the excellent potential of PV installation for the future investments of RE projects in Dongola, and generally in Sudan as the vital options and best alternate for conventional systems.

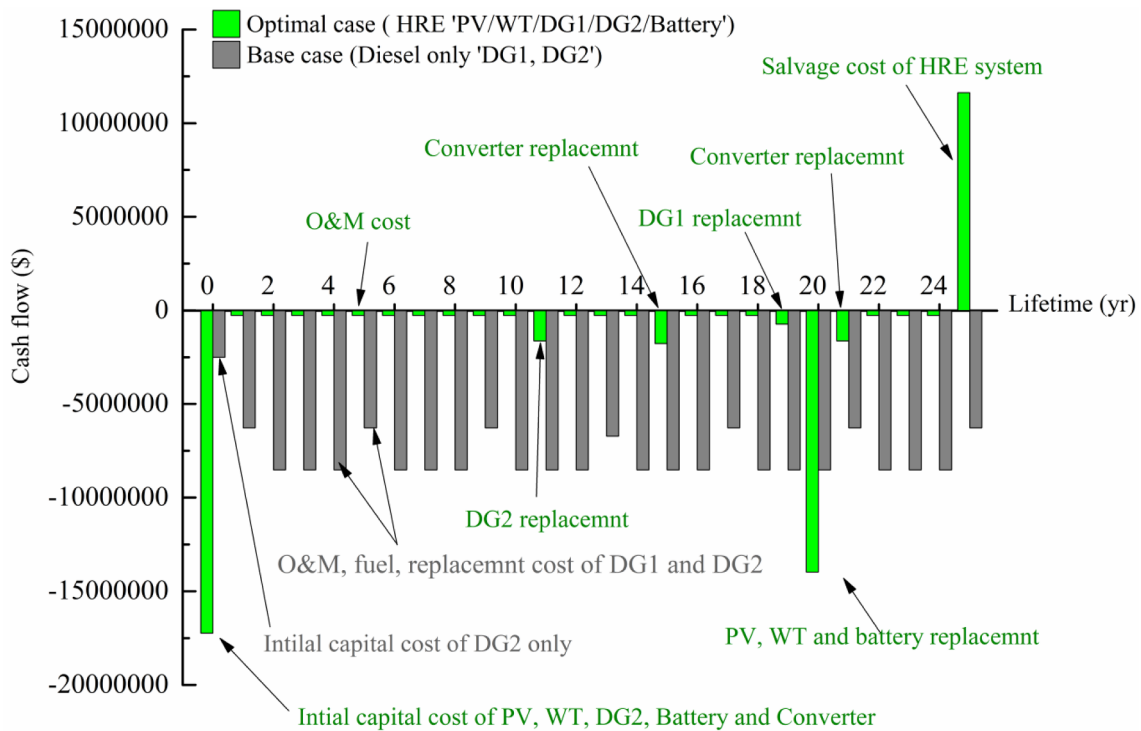


Fig. 15. Annual nominal cash flow of the optimal configuration plan compared to the base case.

For more illustration, Fig. 18 presents how the total system costs distributed between system components.

4.2.3. Environmental performance

The further analysis of the environmental impact of the optimal HRES showed that the resultant GHG emission was reduced significantly from 351, 3972 kg/yr with the base case (zero RF) to just 168,786 owing to the high inclusion of RE (95.12% RF). This reduction justifies eliminating the ineffective utilization of diesel generators and results in decreasing the dependency on fuel-fossil. As demonstrated above in Table 3, the total number of operating hours for both DG1 and DG2 was reduced extremely from 9223 hr/yr to only 349 hr/yr. Therefore, the total amount of consumed diesel fuel was obviously decreased to around 36,798 L/yr for optimal case compared to 1,328,214 L/yr for the base case. As a result, this will increase the sustainability of the proposed optimal HRES.

Table 5

Summary of electrical energy production and consumption results for optimal configuration plan.

Source	(kWh/yr)	%	Load	kWh/yr	%
PV	3,820,438	63.1	Electrical load	4,883,02	100
WT	1,997,016	33	Electrical excess	584,545	9.65
DG1	36,521	0.603	Unmet load	677	0.0139
DG2	201,567	3.33	Capacity Shortage	1,587	0.0325
Total production	6,055,541	100			

4.3. Sensitivity analysis

After the HRES is designed and optimized subject to load requirements and considering certain resources and economic conditions, different system parameters including renewable resource-based parameters (solar irradiance, wind speed) and economic parameters (fuel price, component costs, interest rate) would fluctuate based on the accuracy of data collection, time changes and policy regulations [81].

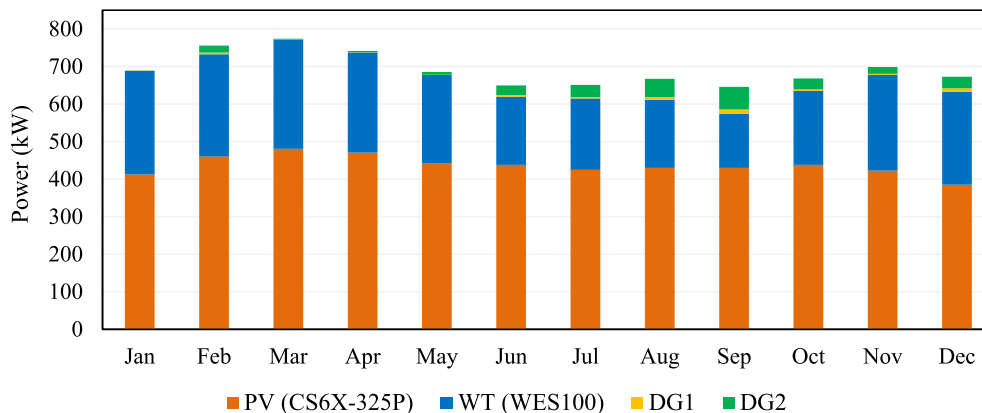


Fig. 16. Monthly average electric production with the optimal configuration plan.

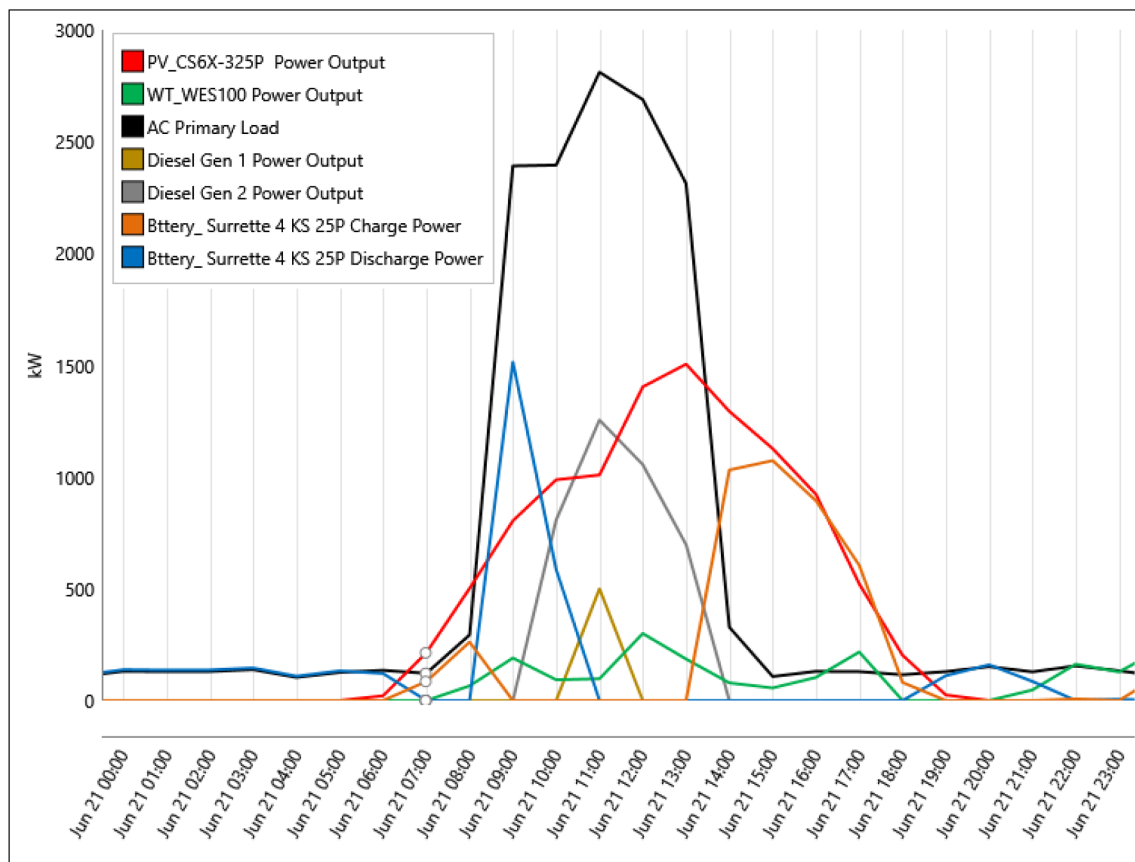


Fig. 17. Hourly power dispatch of the optimal configuration plan, (sample day, 21 June).

Consequently, this would impact on the actual system design and operation. Therefore, it is inevitable for performing sensitivity analysis.

In general, sensitivity analysis (or what-if analysis) is applied to identify the key input parameters influencing the energy system [33]. In other words, sensitivity analysis aims to quantify the importance of input parameters regarding their contribution to drastic changes in model output variability. In this study, the sensitivity analysis was worked out by taking the effect changes in four sensitivity parameters, namely average solar radiation, average wind speed, diesel fuel price, and interest rate. The study was conducted considering a realistic variation of average solar radiation and average wind speed in the range from -20% to $+20\%$, while the diesel fuel price values are 0.5, 1.0, 1.28, and 2.0 \$/L. Also, the rate of interest was varied from 4% to 14%. The considered values of the investigated sensitivity parameters are presented in Table 7. Regarding the uncertainty of solar radiation and wind speed, it should mention that the sensitivity analysis was appropriately conducted on the hourly data sets. Where the new hourly data sets at each scaled annual average value were obtained by scaling the baseline hourly data sets defined at best estimate values of annual average solar radiation ($6.2/\text{kWh}/\text{m}^2$) and wind speed (4.21 m/s). The

corresponding scaling factor for each hourly data set was obtained by dividing the scaled annual average value by the baseline annual average value. As a result, the uncertainty related to high-resolution profiles of the renewable energy resources was properly considered rather than the average value.

Because of the solar radiation and wind speed data are normally acquired from satellite estimation over the project site. But such data must be correctly obtained from the actual weather data recorded over a long period through a site survey. Also, the optimization analysis was worked out considering certain hourly data sets for solar radiation and wind speed. However, for more accurate design, it is essential to model and analyse the impact a good (windy sunny) and worse (unwind and with a cloudy sky) day conditions on the system optimization. Considering that, the variability range of the average solar radiation and average wind speed of a certain year is considered within range between -20% (as the worst case) and $+20\%$ (as the best optimistic case) as in [22] and [50]. On the other side, the simulation was performed considering a fixed value for diesel fuel price. But this value is expected to increase through the project lifetime; for instant, if the government reduce the subsidy on the diesel fuel prices, the fuel price will be increased in the future. Also, the frequent changes in global buying and selling prices of diesel fuel and how much of the world's fuel production and demand changes may be effect on the cost of the diesel fuel. Therefore, to cover the future increase and decrease of the diesel fuel price, the sensitivity analysis was performed at values of 0.5, 1.0, 1.28, and 2.0 \$/L. Finally, according to Trading Economic [82], the interest rate in Sudan has changed from 4 to around 14% since 2003. So, it is essential to investigate this considerable fluctuation of the interest rate on the financial performance of the recommended HRES for Dongola.

The flowchart applied for sensitivity analysis is illustrated in Fig. 19. Firstly, the sensitivity factors were defined to HOMER Pro® with the

Table 6

Cost summary of the optimal configuration plan.

Cost (M\$) Component	Capital	Replacement	O&M	Fuel	Salvage	Total
PV	1.400	0.437	0.358	0.000	-0.245	1.950
WT	8.400	2.619	0.081	0.000	-1.468	9.632
DG1	0.000	0.151	0.319	0.160	-0.070	0.561
DG2	1.500	1.131	1.713	0.883	-0.193	5.034
Battery	4.439	1.218	0.000	0.000	-0.683	4.974
Converter	1.497	0.625	0.000	0.000	-0.116	2.005
System	17.236	6.180	2.470	1.044	-2.775	24.155

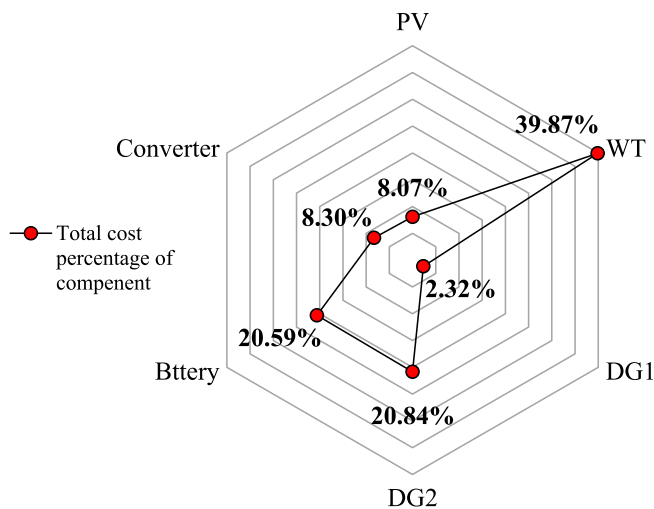


Fig. 18. Break down of the total cost by component of the optimal configuration plan.

data mentioned in Table 7. Using scenario based-technique, a number of sensitivity cases equal to 144 ($3 \times 3 \times 4 \times 4$) were created, e.g. the first scenario will have solar radiation equals 5.024, wind speed equals 3.368 m/s, diesel price equals 0.5 \$/L and 4% interest rate. In the second one, solar radiation is 6.28, wind speed equals 3.368 m/s, diesel price equals 0.5 \$/L, and 4% interest rate, and so on. With each sensitivity case, system simulation and optimization were performed again, and a set of optimized configurations was generated. The winning configuration with each sensitivity case was the one that achieves the minimum value of NPC in most feasible configurations.

As depicted from the graph result in Fig. 20, the winning HRES for Dongola seems to rely more on PV-WT-DG1-DG2-battery-converter hybrid configuration even with the variations of the defined sensitivity parameters, where this configuration was selected to be the winning solution in most cases and marked by the blue area. This shows the superior economic performance of the recommended HRES over the conventional energy system for Dongola, which implies the system is sustainable to the fluctuations of RE resource and the change of diesel fuel price and the interest rate increase.

To analyze and evaluate how uncertain parameters influence the economic performance of the HRES, a spider graph is presented in Fig. 21. The graph reveals the response of NPC and COE with the variation the considered sensitivity variables. Notably, the fluctuation in solar radiation and wind speed have a considerable impact on NPC and COE. A negative change of 20% in solar radiation and wind speed cause a significant increase in NPC by around 25.85% and 38.27%, respectively. This high increase was accounted due to increasing the number of operating hours and fuel consumption rate for the diesel generators to offset the shortage in the output power of PV and WT in such situations. Likewise, when solar radiation and wind speed are positively varied, both NPC and COE were reduced. The result shows that the NPC and COE were declined from 24.16 M\$ and 0.387 \$/kWh at best estimate values for solar radiation and wind speed to respectively 19.5 M\$ and 0.313 \$/kWh with 20% increase in solar radiation ($7.536 \text{ kWh/m}^2/\text{day}$). Also, the values of NPC and COE were reduced to 19.3 M\$ and

0.309 \$/kWh when the wind speed was increased up to 5.024 m/s.

What stands out also from Fig. 21 is that the variation of interest rate affects significantly the NPC and the COE. As the interest rate rises, the system NPC and capital cost decrease as a result of decreasing the number of WTs and batteries as given in Table 8. Where the minimum NPC of 19.6 M\$ was belonged to the case with interest rate of 14%, achieving a reduction by about 19% than the corresponding value at the baseline case of 6% interest rate. This change is mostly caused due to the higher initial cost and replacement costs of WTs and batteries packs that lead the system to reduce them by increasing the value of the interest rate.

In contrast, the trend in the COE variation moves toward higher value in response to the increase in the interest rate. For example, it was observed that at an interest of 14%, the COE significantly grew to 0.583 \$/kWh with almost 50% increase compared to the corresponding value at the baseline case with 6% interest rate. This attributed to the more substantial growth in the O&M cost caused by increasing the fuel consumption and operating hours of the diesel generators to compensate the decrement of the number of WTs. On the other hand, the variation of the NPC and the COE was very small as the diesel fuel price increase, even with ascending the fuel price to its most unoptimistic value (2.0 \$/L), both NPC and COE change only by around 2.2% and 2.3%, respectively.

Conclusively, the main findings of the conducted sensitivity analysis are that the variations of climate conditions and increase of the interest rate have the major impact on the economic performance of the system, so these parameters should be taken into account by project managers during investment decision stage. Meanwhile, the effect of changes in the price of diesel fuel is lowly regarded for future investors of renewable energy projects in Sudan. The main concern with diesel's investment decision would be to ensure high system reliability under the constraints of the harmful GHG emissions.

4.4. Comparison with economic results presented in the literature

However it was difficult to compare the present case study with those listed in the literature due to the variation of the hybrid energy structure, component sizes, and load requirements, the system economic metrics were considered as appropriate criteria to evaluate and compare the obtained results in this work with those values available in the literature. Table 9 provides a synoptic comparison between NPC and COE of the current study of Dongola, Sudan and those values of different HRES plans installed in various locations all over the world.

Table 7
Studied values of sensitivity parameters.

Sensitivity Parameter	Range	Scaled values
Solar radiation ($\text{kWh/m}^2/\text{day}$)	$\pm 20\%$	5.024, 6.28, 7.536
Wind speed (m/s)	$\pm 20\%$	3.368, 4.21, 5.052
Diesel fuel price (\$/L)	–	0.5, 1.0, 1.28, 2.0
Interest rate (%)	–	4, 6, 10, 14

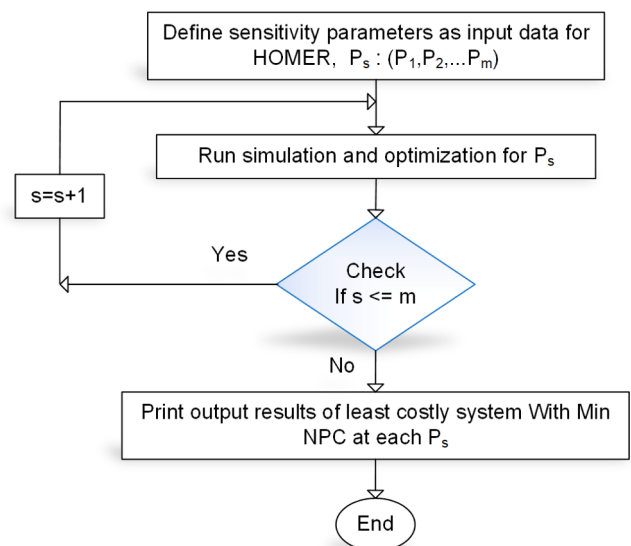


Fig. 19. Method applied for sensitivity analysis.

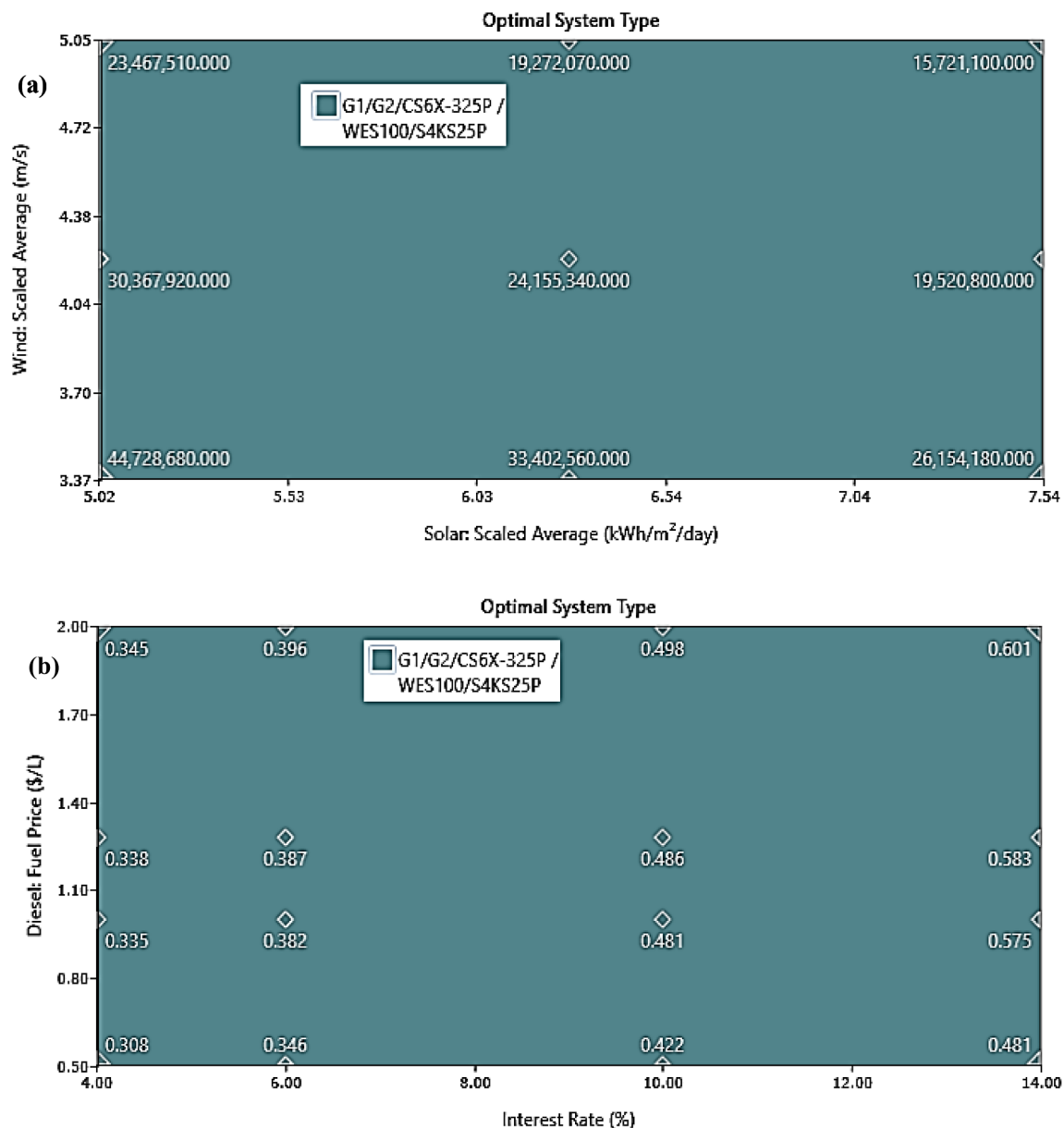


Fig. 20. Graphical representation showing optimal system type with the variation of sensitivity parameters: (a) NPC with the variation of solar radiation and wind speed at a fixed value for diesel fuel price (1.28 \$/L) and interest rate (6%), (b) COE with the variation of diesel fuel price and interest rate at fixed solar radiation (6.281 kWh/m²/day) wind speed (4.21 m/s).

Because of the total NPC has been determined on the basis of sizes and local capital costs of HRES components, its value was meaningfully unequal and differed from region to another. However, the values of COE can be used instead as a valuable indicator of the cost of RE production. From the tabulated survey, it is witnessed that Shafar (Yemen) and Hurghada (Egypt) had the least COE in comparison with other regions. In contrast, the energy cost of HRES in Australia was higher than in the other countries. According to the obtained value of COE of the current study, this comparison shows a good agreement with the other works and can give insight concerning the economic viability of HRES in the Sudan, and in the global as well.

5. Conclusions

In this paper, the feasibility of grid-isolated HRES, which aims at electrification of an agriculture and irrigation area in Dongola, Sudan has been investigated. A proposed integrated and systematic framework for optimal planning and design of HRES has been introduced. The

framework is based on techno-economic simulation and optimization analysis for different HRES configurations with varying levels of penetration of RESs. The study analyses and assess different performance metrics through a comparative study of the various configurations to find the winning solution at least NPC and reasonable GHG emissions. Sensitivity analysis has also been executed to outline the impact of the uncertainty of design parameters on HRES design and cost.

The obtained simulation results could lead to conclusions highlighted as follows:

- The comparison of the different cases reveals that the hybridization of 2000 kW PV, 42 WTs (100 kW rating), 1500 kW DG1, 3551 batteries (7.55 kWh nominal capacity) and 2907 kW power converter, in addition to the existing DG1 of 500 kW rating shows the optimal techno-economic performance among all investigated cases.
- The optimal system brings down the NPC and COE to its minimum values of 24.16 M\$ and 0.387 \$/kWh, respectively, as well as guarantee the practicability of high penetration of renewables,

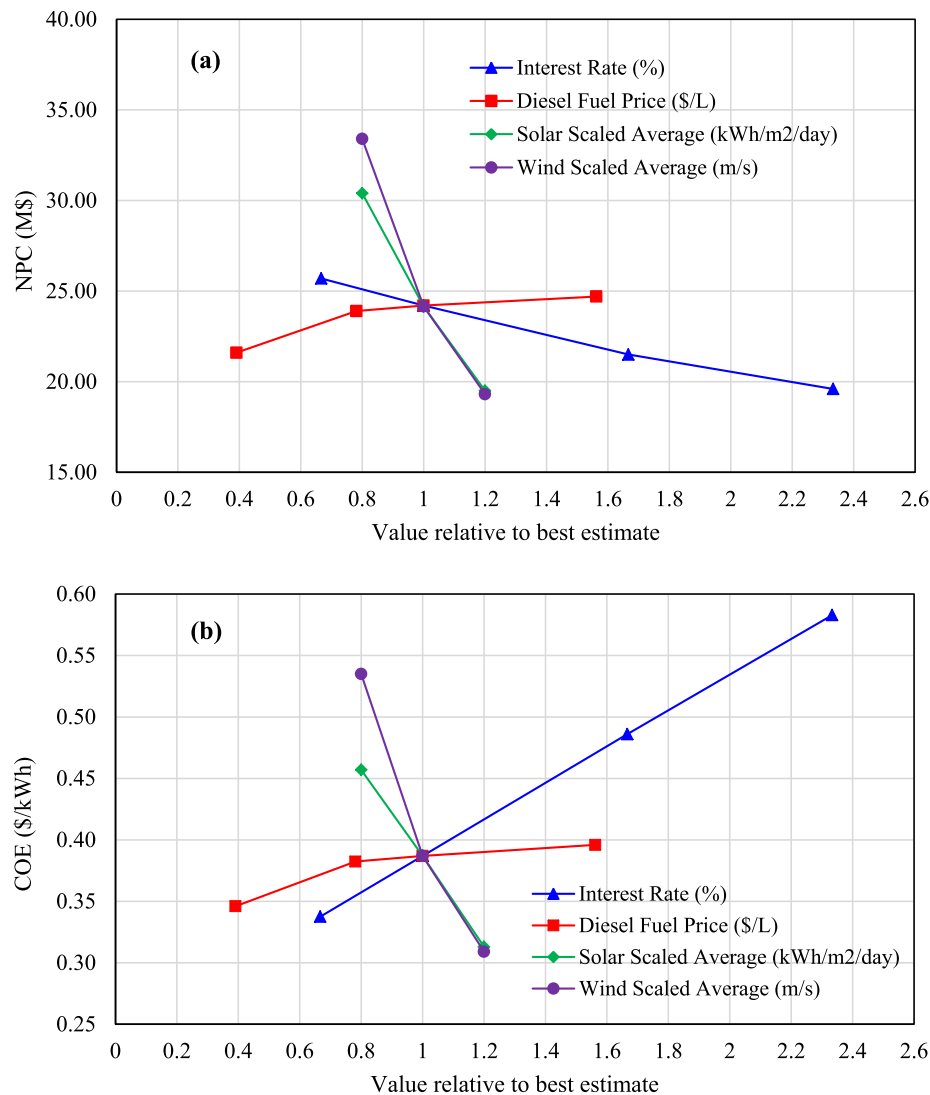


Fig. 21. Effect of sensitivity variation on the economic performance of the designed HRES: (a) Effect on NPC, (b) Effect on COE. (Best estimate values: solar radiation 6.28 kWh/m²/day, wind speed 4.21 m/s, diesel fuel price 1.28 \$/L and interest rate 6%).

Table 8

Cost-optimal system configuration at different interest rates.

Interest rate (%)	PV (kW)	WT (Qty.)	DG1 (kW)	DG2 (kW)	Battery (Qty.)	Converter (kW)	NPC (M \$)	COE (\$/kWh)	Capital cost (M\$)	O&M cost (\$/yr)	RF (%)	Fuel consumption (L/yr)
4	2000	45	500	1500	3720	2889	25.7	0.338	18.0	493,59	95.9	53,598
6	2000	42	500	1500	3551	2907	24.2	0.387	17.2	541,29	95.1	63,798
10	2000	37	500	1500	3161	2763	21.5	0.486	15.7	646,67	93.2	89,295
14	2000	28	500	1500	2533	2628	19.6	0.583	13.0	951,59	89.1	143,12

particularly PV with RF equals to 95.12%. Additionally, a considerable value of the financial return based on ROI of 39.94% was achieved. Thereby, the system has been presented as cost-effective and almost near the most sustainable configuration for supplying the investigated area.

- Meanwhile, several benefits have been obtained with the inclusion of RE resources including; supporting hundred percent of the load demand with excess energy of 9.65% and reducing the harmful GHG emissions and fuel consumption by almost 95% compared with the base case using DGs only.
- Furthermore, the sensitivity analysis results show the trend towards the high utilization of RESs and less dependency on DGs under the variation of RE resources, diesel fuel price, and interest rate.

The study highlights that the NPC and COE of proposed HRES are more dependent on the variations of the RE resources (i.e., solar radiation and wind speed) and the interest rate, together with the possibility of ignoring the effect of changes in diesel fuel price in the investment decision.

Conclusively, the proposed HRES can attain the national targets of the Sudan government's plan and the international aims of UN and Paris HCA toward clean and sustainable energy supply. It is also expected that the developed framework with the presented case study can be used as a valuable benchmark to assist energy system planners and decision-makers in identifying the optimal adoption and integration of HRESs for other locations with the availability of similar RE resources.

Table 9
NPC and COE of the optimal HRES configuration plan in different regions all over the world.

Region	Province/City/village	HRES plan	NPC (M\$)	COE (\$/kWh)	Ref
Egypt	Saint Catherine,	PV/WT/FC/DG/Natural gas turbine/ Biomass generator /Battery/Converter	0.165	0.532	[21]
Yemen	Shafar	PV/ WT/ DG/Battery/Converter	0.722	0.137	[22]
Egypt	Hurghada	PV/ WT/ DG/Battery/Converter	0.822	0.139	[23]
Somalia	Hargeisa	PV/ WT/Battery/Converter	1.72	0.288	[26]
India	Silchar	PV/Biogas generator/Pumped hydro energy	0.813	0.4864	[28]
Malaysia	South China Sea	PV/ WT/ DG/Battery/Converter	17.15	0.279	[33]
KSA	Yanbu	PV/WT/ Battery/Converter	0.038	0.609	[37]
Australia	Sydney	PV/WT/Battery/Converter	0.093	1.502	[37]
Colombia	Puerto Estrella	PV/ DG/Battery/Converter	0.836	0.473	[39]
Bangladesh	Chorasariadho,	PV/ WT/ DG/Battery/Converter	0.335	0.37	[40]
KSA	Rafha,	PV/ DG/Battery/Converter	28.5	0.170	[42]
India	Amritsar,	PV/ WT/ DG/Battery/Converter	0.010	0.164	[44]
Canary Islands	Lanzarote,	PV/ DG/Battery/Converter	0.473	0.404	[46]
Malaysia	Pulau Banggi	PV/ WT/ DG/Battery/Converter	8.54	0.276	[47]
South Korea	Busan,	PV/WT/Battery/Converter	26.09	0.399	[56]
Canada	Vancouver	PV/DG/Natural gas generator/Biomass/Battery/Converter	29.3	0.285	[57]
Sudan	Dongola	PV/WT/DG/Battery/Converter	24.16	0.387	Current study

and load conditions. For future research directions, apart from the technical and economic objectives deliberated in this study, further studies incorporating multi-objective optimization can also be conducted by integrating other objectives and constraints associated with environmental and socio-political perspectives. This expected to enhance and precision and the practical significance of the HRES planning and design.

Declaration of Competing Interest

The authors declare that there is no competing interest.

References

- [1] Indragandhi V, Subramaniaswamy V, Logesh R. Resources, configurations, and soft computing techniques for power management and control of PV/wind hybrid system. *Renew Sustain Energy Rev* 2017;69:129–43. <https://doi.org/10.1016/j.rser.2016.11.209>.
- [2] IRENA. Global Energy, a Roadmap to 2050 by IRENA. 2018 < <http://www.irena.org> > .
- [3] Dawoud SM, Lin X, Okba MI. Hybrid renewable microgrid optimization techniques: a review. *Renew Sustain Energy Rev* 2018;82:2039–52. <https://doi.org/10.1016/j.rser.2017.08.007>.
- [4] Djelailia O, Kelaiaia MS, Labar H, Necaibia S, Merad F. Energy hybridization photovoltaic/diesel generator/pump storage hydroelectric management based on on-line optimal fuel consumption per kWh. *Sustain Cities Soc* 2019;44:1–15. <https://doi.org/10.1016/j.scs.2018.09.037>.
- [5] Quek TYA, Alvin Ee WL, Chen W, Ng TSA. Environmental impacts of transitioning to renewable electricity for Singapore and the surrounding region: A life cycle assessment. *J Clean Prod* 2019;214:1–11. <https://doi.org/10.1016/j.jclepro.2018.12.263>.
- [6] Zhang L, Xin H, Yong H, Kan Z. Renewable energy project performance evaluation using a hybrid multi-criteria decision-making approach: case study in Fujian, China. *J Clean Prod* 2019;206:1123–37. <https://doi.org/10.1016/j.jclepro.2018.09.059>.
- [7] Shouman ER. International and national renewable energy for electricity with optimal cost effective for electricity in Egypt. *Renew Sustain Energy Rev* 2017;77:916–23. <https://doi.org/10.1016/j.rser.2016.12.107>.
- [8] Martin-Martínez F, Sánchez-Mirallas A, Rivier M. A literature review of microgrids: a functional layer based classification. *Renew Sustain Energy Rev* 2016;62:1133–53. <https://doi.org/10.1016/j.rser.2016.05.025>.
- [9] Kosai S. Dynamic vulnerability in standalone hybrid renewable energy system. *Energy Convers Manag* 2019;180:258–68. <https://doi.org/10.1016/j.enconman.2018.10.087>.
- [10] Genwa KR, Sagar CP. Energy efficiency, solar energy conversion and storage in photogalvanic cell. *Energy Convers Manag* 2013;66:121–6. <https://doi.org/10.1016/j.enconman.2012.10.007>.
- [11] Frankfurt School-UNEP Centre/BNEF. Global Trends in Renewable Energy Investment. 2018 < <https://europa.eu/capacity4dev/unep/documents/global-trends-renewable-energy-investment-2018> > .
- [12] Zia MF, Elbouchikhi E, Benbouzid M. Microgrids energy management systems: a critical review on methods, solutions, and prospects. *Appl Energy* 2018;222:1033–55. <https://doi.org/10.1016/j.apenergy.2018.04.103>.
- [13] Suman S. Hybrid nuclear-renewable energy systems: a review. *J Clean Prod* 2018;181:166–77. <https://doi.org/10.1016/j.jclepro.2018.01.262>.
- [14] Jordehi AR. How to deal with uncertainties in electric power systems? A review. *Renew Sustain Energy Rev* 2018;96:145–55. <https://doi.org/10.1016/j.rser.2018.07.056>.
- [15] Khodadoost Arani AA, Gharehpetian BG, Abedi M. Review on energy storage systems control methods in microgrids. *Int J Electr Power Energy Syst* 2019;107:745–57. <https://doi.org/10.1016/j.ijepes.2018.12.040>.
- [16] Bahramara S, Moghaddam MP, Haghifam MR. Optimal planning of hybrid renewable energy systems using HOMER: a review. *Renew Sustain Energy Rev* 2016;62:609–20. <https://doi.org/10.1016/j.rser.2016.05.039>.
- [17] Siddaiah R, Saini RP. A review on planning, configurations, modeling and optimization techniques of hybrid renewable energy systems for off grid applications. *Renew Sustain Energy Rev* 2016;58:376–96. <https://doi.org/10.1016/j.rser.2015.12.281>.
- [18] Twaha S, Ramli MAM. A review of optimization approaches for hybrid distributed energy generation systems: off-grid and grid-connected systems. *Sustain Cities Soc* 2018;41:320–31. <https://doi.org/10.1016/j.scs.2018.05.027>.
- [19] Abdelshafy AM, Hassan H, Jurasz J. Optimal design of a grid-connected desalination plant powered by renewable energy resources using a hybrid PSO–GWO approach. *Energy Convers Manag* 2018;173:331–47. <https://doi.org/10.1016/j.enconman.2018.07.083>.
- [20] Sawle Y, Gupta SC, Bohre AK. Review of hybrid renewable energy systems with comparative analysis of off-grid hybrid system. *Renew Sustain Energy Rev* 2018;81:2217–35. <https://doi.org/10.1016/j.rser.2017.06.033>.
- [21] Abo-elyousr FK, Elnozahy A. Bi-objective economic feasibility of hybrid micro-grid systems with multiple fuel options for islanded areas in Egypt. *Renew Energy* 2018;128:37–56. <https://doi.org/10.1016/j.renene.2018.05.066>.
- [22] Ajlan Abdullah, Tan Chee Wei, Abdilahi AM. Assessment of environmental and economic perspectives for renewable-based hybrid power system in Yemen. *Renew Sustain Energy Rev* 2016;31:2343–7. <https://doi.org/10.1016/j.rser.2016.11.024>.
- [23] Dawoud SM, Lin XN, Sun JW, Okba MI, Khalid MS, Waqar A. Feasibility study of isolated PV-wind hybrid system in Egypt. *Adv Mater Res* 2015;1092–1093:145–51. <https://doi.org/10.4028/www.scientific.net/AMR.1092-1093.145>.
- [24] Sawle Y, Gupta SC, Bohre AK. Socio-techno-economic design of hybrid renewable energy system using optimization techniques. *Renew Energy* 2018;119:459–72. <https://doi.org/10.1016/j.renene.2017.11.058>.
- [25] Ma T, Javed MS. Integrated sizing of hybrid PV-wind-battery system for remote island considering the saturation of each renewable energy resource. *Energy Convers Manag* 2019;182:178–90. <https://doi.org/10.1016/j.enconman.2018.12.059>.
- [26] Abdilahi AM, Mohd Yatim AH, Mustafa MW, Khalaf OT, Shumran AF, Mohamed Nor F. Feasibility study of renewable energy-based microgrid system in Somaliland's urban centers. *Renew Sustain Energy Rev* 2014;40:1048–59. <https://doi.org/10.1016/j.rser.2014.07.150>.
- [27] Bastholm C, Fiedler F. Techno-economic study of the impact of blackouts on the viability of connecting an off-grid PV-diesel hybrid system in Tanzania to the national power grid. *Energy Convers Manag* 2018;171:647–58. <https://doi.org/10.1016/j.enconman.2018.05.107>.
- [28] Das M, Singh MAK, Biswas A. Techno-economic optimization of an off-grid hybrid renewable energy system using metaheuristic optimization approaches – case of a radio transmitter station in India. *Energy Convers Manag* 2019;185:339–52. <https://doi.org/10.1016/j.enconman.2019.01.107>.
- [29] Huang Z, Xie Z, Zhang C, Chan SH, Milewski J, Xie Y, et al. Modeling and multi-objective optimization of a stand-alone PV-hydrogen-retired EV battery hybrid energy system. *Energy Convers Manag* 2019;181:80–92. <https://doi.org/10.1016/j.enconman.2018.11.079>.
- [30] Duman AC, Güler Ö. Techno-economic analysis of off-grid PV/wind/fuel cell hybrid system combinations with a comparison of regularly and seasonally occupied households. *Sustain Cities Soc* 2018;42:107–26. <https://doi.org/10.1016/j.scs.2018.06.029>.

- [31] Luta DN, Raji AK. Decision-making between a grid extension and a rural renewable off-grid system with hydrogen generation. *Int J Hydrogen Energy* 2018;43:9535–48. <https://doi.org/10.1016/j.ijhydene.2018.04.032>.
- [32] Baneshi M, Hadianfar F. Techno-economic feasibility of hybrid diesel/PV/wind/battery electricity generation systems for non-residential large electricity consumers under southern Iran climate conditions. *Energy Convers Manag* 2016;127:233–44. <https://doi.org/10.1016/j.enconman.2016.09.008>.
- [33] Hossain M, Mekhilef S, Olatomiwa L. Performance evaluation of a stand-alone PV-wind-diesel-battery hybrid system feasible for a large resort center in South China Sea. *Malaysia. Sustain Cities Soc* 2017;28:358–66. <https://doi.org/10.1016/j.scs.2016.10.008>.
- [34] Singh A, Baredar P, Gupta B. Computational simulation & optimization of a solar, fuel cell and biomass hybrid energy system using HOMER pro software. *Procedia Eng.* 2015;127:743–50. <https://doi.org/10.1016/j.proeng.2015.11.408>.
- [35] Rammal R, Koubayssi A. Design and optimization of a hybrid micro grid. 2017 Sensors Networks Smart Emerg Technol SENSET 2017 2017;2017-Janua:1–4. Doi: 10.1109/SENSET.2017.8125022.
- [36] Zahboune H, Ziani E, Elhafyani M, Krajacic G, Varbanov PS, Zouggar S. Optimal hybrid renewable energy design in autonomous system using modified electric system cascade analysis and homer software. *Energy Convers Manag* 2016;126:909–22. <https://doi.org/10.1016/j.enconman.2016.08.061>.
- [37] Al-Sharafi A, Sahin AZ, Ayar T, Yilbas BS. Techno-economic analysis and optimization of solar and wind energy systems for power generation and hydrogen production in Saudi Arabia. *Renew Sustain Energy Rev* 2017;69:33–49. <https://doi.org/10.1016/j.rser.2016.11.157>.
- [38] Dufo-López R, Cristóbal-Monreal IR, Yusta JM. Optimisation of PV-wind-diesel-battery stand-alone systems to minimise cost and maximise human development index and job creation. *Renew Energy* 2016;94:280–93. <https://doi.org/10.1016/j.renene.2016.03.065>.
- [39] Haghighat Mamaghani A, Avella Escandon SA, Najafi B, Shirazi A, Rinaldi F. Techno-economic feasibility of photovoltaic, wind, diesel and hybrid electrification systems for off-grid rural electrification in Colombia. *Renew Energy* 2016;97:293–305. <https://doi.org/10.1016/j.renene.2016.05.086>.
- [40] Mandal S, Das BK, Hoque N. Optimum sizing of a stand-alone hybrid energy system for rural electrification in Bangladesh. *J Clean Prod* 2018;200:12–27. <https://doi.org/10.1016/j.jclepro.2018.07.257>.
- [41] Jung J, Villaran M. Optimal planning and design of hybrid renewable energy systems for microgrids. *Renew Sustain Energy Rev* 2017;75:180–91. <https://doi.org/10.1016/j.rser.2016.10.061>.
- [42] Shaahid SM, El-Amin I. Techno-economic evaluation of off-grid hybrid photovoltaic-diesel-battery power systems for rural electrification in Saudi Arabia-A way forward for sustainable development. *Renew Sustain Energy Rev* 2009;13:625–33. <https://doi.org/10.1016/j.rser.2007.11.017>.
- [43] Akinyele D. Techno-economic design and performance analysis of nanogrid systems for households in energy-poor villages. *Sustain Cities Soc* 2017;34:335–57. <https://doi.org/10.1016/j.scs.2017.07.004>.
- [44] Khan MJ, Yadav AK, Mathew L. Techno economic feasibility analysis of different combinations of PV-Wind-Diesel-Battery hybrid system for telecommunication applications in different cities of Punjab, India. *Renew Sustain Energy Rev* 2017;76:577–607. <https://doi.org/10.1016/j.rser.2017.03.076>.
- [45] Kanase-Patil AB, Saini RP, Sharma MP. Integrated renewable energy systems for off grid rural electrification of remote area. *Renew Energy* 2010;35:1342–9. <https://doi.org/10.1016/j.renene.2009.10.005>.
- [46] Padrón I, Avila D, Marichal GN, Rodríguez JA. Assessment of hybrid renewable energy systems to supplied energy to autonomous desalination systems in two islands of the canary archipelago. *Renew Sustain Energy Rev* 2019;101:221–30. <https://doi.org/10.1016/j.rser.2018.11.009>.
- [47] Halabi LM, Mekhilef S, Olatomiwa L, Hazelton J. Performance analysis of hybrid PV/diesel/battery system using HOMER: a case study Sabah. *Malaysia. Energy Convers Manag* 2017;144:322–39. <https://doi.org/10.1016/j.enconman.2017.04.070>.
- [48] Baghdadi F, Mohammedi K, Diaf S, Behar O. Feasibility study and energy conversion analysis of stand-alone hybrid renewable energy system. *Energy Convers Manag* 2015. <https://doi.org/10.1016/j.enconman.2015.07.051>.
- [49] Ramli MAM, Boucekara HREH, Alghamdi AS. Optimal sizing of PV/wind/diesel hybrid microgrid system using multi-objective self-adaptive differential evolution algorithm. *Renew Energy* 2018;121:400–11. <https://doi.org/10.1016/j.renene.2018.01.058>.
- [50] Das HS, Tan CW, Yatim AHM, Lau KY. Feasibility analysis of hybrid photovoltaic/battery/fuel cell energy system for an indigenous residence in East Malaysia. *Renew Sustain Energy Rev* 2017;76:1332–47. <https://doi.org/10.1016/j.rser.2017.01.174>.
- [51] Gan LK, Shek JKH, Mueller MA. Hybrid wind-photovoltaic-diesel-battery system sizing tool development using empirical approach, life-cycle cost and performance analysis: a case study in Scotland. *Energy Convers Manag* 2015;106:479–94. <https://doi.org/10.1016/j.enconman.2015.09.029>.
- [52] Shahzad MK, Zahid A, Rashid T, Rehan MA, Ali M, Ahmad M. Techno-economic feasibility analysis of a solar-biomass off grid system for the electrification of remote rural areas in Pakistan using HOMER software. *Renew Energy* 2017;106:264–73. <https://doi.org/10.1016/j.renene.2017.01.033>.
- [53] Shezan SA, Julai S, Kibria MA, Ullah KR, Saidur R, Chong WT, et al. Performance analysis of an off-grid wind-PV (photovoltaic)-diesel-battery hybrid energy system feasible for remote areas. *J Clean Prod* 2016;125:121–32. <https://doi.org/10.1016/j.jclepro.2016.03.014>.
- [54] Jamshidi M, Askarzadeh A. Techno-economic analysis and size optimization of an off-grid hybrid photovoltaic, fuel cell and diesel generator system. *Sustain Cities Soc* 2019;44:310–20. <https://doi.org/10.1016/j.scs.2018.10.021>.
- [55] Maleki A, Khajeh MG, Ameri M. Optimal sizing of a grid independent hybrid renewable energy system incorporating resource uncertainty, and load uncertainty. *Int J Electr Power Energy Syst* 2016;83:514–24. <https://doi.org/10.1016/j.ijepes.2016.04.008>.
- [56] Bagheri M, Delbari SH, Pakzadmanesh M, Kennedy CA. City-integrated renewable energy design for low-carbon and climate-resilient communities. *Appl Energy* 2019;239:1212–25. <https://doi.org/10.1016/j.apenergy.2019.02.031>.
- [57] Baek S, Park E, Kim MG, Kwon SJ, Kim KJ, Ohm JY, et al. Optimal renewable power generation systems for Busan metropolitan city in South Korea. *Renew Energy* 2016;88:517–25. <https://doi.org/10.1016/j.renene.2015.11.058>.
- [58] Zhang Q, Wang G, Li Y, Li H, McLellan B, Chen S. Substitution effect of renewable portfolio standards and renewable energy certificate trading for feed-in tariff. *Appl Energy* 2018;227:426–35. <https://doi.org/10.1016/j.apenergy.2017.07.118>.
- [59] Bagheri M, Shirzadi N, Bazdar E, Kennedy CA. Optimal planning of hybrid renewable energy infrastructure for urban sustainability: green Vancouver. *Renew Sustain Energy Rev* 2018;95:254–64. <https://doi.org/10.1016/j.rser.2018.07.037>.
- [60] Eriksson ELV, Gray EMA. Optimization and integration of hybrid renewable energy hydrogen fuel cell energy systems – a critical review. *Appl Energy* 2017;202:348–64. <https://doi.org/10.1016/j.apenergy.2017.03.132>.
- [61] HOMER Pro® Version 3.12 User Manual. 2018 < <https://www.homerenergy.com/products/pro/docs/3.12/index.html> > .
- [62] Mariam L, Basu M, Conlon MF. Microgrid: architecture, policy and future trends. *Renew Sustain Energy Rev* 2016;64:477–89. <https://doi.org/10.1016/j.rser.2016.06.037>.
- [63] Abedi S, Alimardani A, Gharehpetian GB, Riahy GH, Hosseini SH. A comprehensive method for optimal power management and design of hybrid RES-based autonomous energy systems. *Renew Sustain Energy Rev* 2012;16:1577–87. <https://doi.org/10.1016/j.rser.2011.11.030>.
- [64] Ullah Z, Wang S, Radosavljevic J, Lai J. A solution to the optimal power flow problem considering WT and PV generation. *IEEE Access* 2019;7:46763–72. <https://doi.org/10.1109/ACCESS.2019.2909561>.
- [65] Acuña LG, Lake M, Padilla RV, Lim YY, Ponzón EG, Soo Too YC. Modelling autonomous hybrid photovoltaic-wind energy systems under a new reliability approach. *Energy Convers Manag* 2018;172:357–69. <https://doi.org/10.1016/j.enconman.2018.07.025>.
- [66] Isa NM, Das HS, Tan CW, Yatim AHM, Lau KY. A techno-economic assessment of a combined heat and power photovoltaic/fuel cell/battery energy system in Malaysia hospital. *Energy* 2016;112:75–90. <https://doi.org/10.1016/j.energy.2016.06.056>.
- [67] Al Garni HZ, Awasthi A, Ramli MAM. Optimal design and analysis of grid-connected photovoltaic under different tracking systems using HOMER. *Energy Convers Manag* 2018;155:42–57. <https://doi.org/10.1016/j.enconman.2017.10.090>.
- [68] Halabi LM, Mekhilef S. Flexible hybrid renewable energy system design for a typical remote village located in tropical climate. *J Clean Prod* 2018;177:908–24. <https://doi.org/10.1016/j.jclepro.2017.12.248>.
- [69] Sinha S, Chandel SS. Review of software tools for hybrid renewable energy systems. *Renew Sustain Energy Rev* 2014;32:192–205. <https://doi.org/10.1016/j.rser.2014.01.035>.
- [70] Pan Y, Liu L, Zhu T, Zhang T, Zhang J. Feasibility analysis on distributed energy system of Chongming County based on RET Screen software. *Energy* 2017;130:298–306. <https://doi.org/10.1016/j.energy.2017.04.082>.
- [71] Mills A, Al-Hallaj S. Simulation of hydrogen-based hybrid systems using Hybrid2. *Int J Hydrogen Energy* 2004;29:991–9. <https://doi.org/10.1016/j.ijhydene.2004.01.004>.
- [72] Zubi G, Dufo-López R, Pasaoglu G, Pardo N. Techno-economic assessment of an off-grid PV system for developing regions to provide electricity for basic domestic needs: a 2020–2040 scenario. *Appl Energy* 2016;176:309–19. <https://doi.org/10.1016/j.apenergy.2016.05.022>.
- [73] HOMER Energy. 2019 < <https://www.homerenergy.com> > .
- [74] Lambert T, Gilman P, Lilienthal P. Micropower System Modeling with Homer. *Integr Altern Sources Energy* 2006:379–418. <https://doi.org/10.1002/047155621.ch15>.
- [75] International Renewable Energy Agency. Renewable energy in the Arab Region: Overview of developments 2016. Doi: 978-92-95111-09-7.
- [76] World Bank Group. < <https://globalwindatlas.info/downloads/Sudan> > .
- [77] World Bank Group. < <https://globalsolaratlas.info/downloads/Sudan> > .
- [78] IRENA. Pan-Arab Renewable Energy Strategy 2030: Roadmap of Actions for Implementation. 2014. < <https://www.irena.org/publications/2014/Jun/Pan-Arab-Renewable-Energy-Strategy-2030-Roadmap-of-Actions-for-Implementation> > .
- [79] International Energy Agency. WEO-2017 Special Report: Energy Access Outlook. 2017 < <https://webstore.iea.org/weo-2017-special-report-energy-access-outlook> > .
- [80] NASA Prediction of Worldwide Energy Resources. 2019 < <https://www.nasa.gov/> > .
- [81] Ma W, Xue X, Liu G, Zhou R. Techno-economic evaluation of a community-based hybrid renewable energy system considering site-specific nature. *Energy Convers Manag* 2018;171:1737–48. <https://doi.org/10.1016/j.enconman.2018.06.109>.
- [82] Trading Economics. < <http://tradingeconomic.com> > .